MEMORIA CIENTÍFICO-TÉCNICA PROYECTOS COORDINADOS
Convocatoria 2022 - «Proyectos de Generación de Conocimiento»

1. PROJECT DATA

TÍTULO DEL PROYECTO COORDINADO (ACRÓNIMO): Procesos sostenibles ene**R**géticamente en la industria química: c**A**ptura y uso de CO_2 a**S**istidos por calenta**MI**ento microond**A**s. **(RASMIA)**

TITLE OF THE COORDINATED PROJECT (ACRONYM): Towards sustainable ene**R**gy processes in chemical industry: development of CO2 c**A**pture and utilization as**S**isted by **MI**crowave he**A**ting **(RASMIA)**

DATOS DE LOS SUBPROYECTOS - *SUBPROJECTS DATA*

SUBPROYECTO 1:
IP 1 COORDINADOR 1 : Reyes Mallada Viana
IP 2 COORDINADOR 2 : José Luis Hueso Martos
TÍTULO - *TITLE*: Towards sustainable ene**R**gy processes in chemical industry: development of CO_2 c**A**pture and utilization as**S**isted by **MI**crowave he**A**ting **(RASMIA)**

SUBPROYECTO 2:
IP 1 (Nombre y apellidos)**:** Felipe Peñaranda Foix
TÍTULO - *TITLE*: Towards sustainable ene**R**gy processes in chemical industry: development of CO_2 c**A**pture and utilization as**S**isted by **MI**crowave he**A**ting **(RASMIA)**

2. JUSTIFICATION AND CONTRIBUTIONS FROM THE COORDINATION

RASMIA entails the transformation of the energy intensive processes in the chemical industry thanks to the development of microwave heating (MWH) technology. MWH transforms energy from an electromagnetic (EM) wave to heat. In contrast to conventional heating (CH) mechanisms that use slower heat transfer processes via conduction, convection or radiation, microwaves (MW) directly interact with MW susceptor materials and induce a rapid conversion of the electromagnetic energy into heat. This rapid heating provides MWH with distinct features that can be leveraged to increase conversion, selectivity and/or energy efficiency of chemical processes. The final goal is to establish a methodology for designing microwave cavities for adsorption and reaction units, including scale-up.

This objective can only be approached from a multidisciplinary point of view, which includes experts in both disciplines, Chemical Engineering and MW heating. The wavelength of MW ranges from 0.01 m to 1 m and the corresponding frequency range is 0.3–300 GHz. This range corresponds to the telecommunications and radiofrequency (RF); thus, telecommunication engineers are responsible for designing the equipment where these waves propagate. RASMIA involves telecommunication engineers from the Institute of Information and Communication Technologies (ITACA) of the Universitat Politècnica de València (UPV), belonging to the Microwave Division (DiMAS). The coordination of the project is led by researchers from Universidad de Zaragoza (UNIZAR), Instituto de Nanociencia y Materiales de Aragón, INMA, working in the field of Chemical Engineering and Materials Science.

UNIZAR provides expertise in material synthesis and chemical processes including adsorption and catalysis. The research topics of DiMAS group include microwave measurement techniques for evaluation of dielectric properties of materials, microwave heating and numerical techniques for EM modelling. In fact, both teams have collaborated previously, since 2016, and have two publications in common. However, they never participated in a collaborative project before, which is a more powerful scientific coordination scheme. The opportunity of collaboration within the framework of a project with common goals will undoubtedly boost the emergent technology of microwave heating in the chemical industry, benefiting for energy savings as it occurs in other industries, such as food, vulcanization or processing materials where microwaves are implemented. In the case of chemical processes, and in particular in the reactors, a temperature stable operation is a must to avoid runaway

reactions and loose of control. In this sense, **MWH unit for a chemical process can only be designed in close collaboration with experts from both disciplines**.

To guarantee and **effective coordination of the project the following mechanisms** will be implemented:

Governance: The decisions affecting the project objectives and impact will be made by consensus of the PIs of the 2 projects. In case there is no consensus and if it only affects one project, the PI of the project will make the final decision but if it affects both projects, the coordinator will take the decision.

Periodic meetings: (1) Six in person meetings are scheduled, including a kick-off meeting within one month since the confirmation of funding, a mid-term meeting to evaluate the status of the project and discuss about the corresponding reports to funding agency, and last meeting at the end of the project to review main outcomes of the project and taking decisions about intellectual property results. (2) Every two months there will be follow up progress meetings about scientific and financial aspects and corrective actions will be taken in case of critical deviations. These meetings will be online and scheduled in advance through the Microsoft Teams platform, where a team of the project will be created. The presentations of the project will be uploaded to the platform together with relevant documentation of the project.

Short visits to laboratories: This will help on the implementation of equipment and establishing new experimental set-ups: Both teams need the expertise of each other to run the experimental set-ups. The group of UNIZAR will visit UPV for the implementation of the set-up for the reaction unit, providing its expertise in handling these systems and analysis of gas mixtures. On the other hand, UPV will help UNIZAR on the installation of the new MW cavity designed for the project-

Communication: There will be fluid communication among the components of the two teams by email or phone, creating a communication group of the project via WhatsApp or the Microsoft teams platform.

3. JUSTIFICATION AND NOVELTY OF THE PROPOSAL

3.1. Suitability and affinity of the proposal to the objectives and characteristics of the selected modality

Global warming, due to the increase of CO_2 in the atmosphere, is a fact and in recent decades its effects have been demonstrated in major natural disasters associated with climate change. For example, in 2021 intense heat waves in Sacramento (USA) caused major fires, and in the northern region of Rhineland-Westphalia (Germany) there were devastating floods[1]. This global phenomenon needs consensual solutions from different countries, as it was stated in the Paris climate summit in 2015, which set the goal of reducing the global temperature increase by 1.5°C this century. The Paris agreement was signed by 55 countries representing 55% of global emissions[2]. The chemical industry accounts for 30% of the industrial emissions. The goal of achieving net zero global CO_2 emissions by 2050 is a priority objective for the EU in the so-called "European Green Deal"[3].

Figure 1 shows that there is no single strategy for emission reductions but instead a combination of various technologies would make it possible to reach the zero-emission target. In this context, carbon capture, utilisation and storage (CCUS) and increase of electric energy uses in residential, transportation sector and in industrial applications are fundamental. The International Energy Association (IEA) refers to CCUS as a set of technologies that can play an important and diverse role in meeting global energy and climate goals. In particular, CCUS plays a key role in the transition of **energy-intensive industries**, 70% of the emissions are produced by three heavy industries: **steel, cement, and chemicals**. The **cement industry** emitted 2,3Gt of CO_2 in 2019[4], (6.8% of the total) of which 1.4Gt is associated with the processing of the raw material, limestone, and for which **there is no other alternative than CO_2 capture**. Replacing fossil sources with renewable sources, i.e. **decarbonization of**

[1] https://www.nationalgeographic.com/environment/article/this-year-extreme-weather-brought-home-reality-of-climate-change Accesed January 2023.

[2] https://unfccc.int/es/process-and-meetings/the-paris-agreement/que-es-el-acuerdo-de-paris Accessed January 2023.

[3] European University Institute, Freitas, J., Silva, S., Dabrowski, J., *The journey to net zero : 12 perspectives from the frontline of the energy transition in Europe*, European University Institute, 2021, https://data.europa.eu/doi/10.2870/587688 Accessed January 2023.

[4] https://www.globalefficiencyintel.com/new-blog/2021/global-cement-industry-ghg-emissions Accessed January 2023.

the energy-intensive manufacturing industries is one of the most urgent technological challenges for the coming decade. Among those industries, the chemical sector is the most significant energy consumer with 21% of the total consumption in Europe followed by steel industry, both using fuels as primary energy source[5]. The transition scenarios to the low-carbon energy in the chemical process industries are commonly based on the so-called Power-to-X concept, which basically assumes using the **low-carbon or renewable electricity to produce fuels and/or chemicals**. While the electrocatalytic production of fuels from water, CO_2 or nitrogen is undoubtedly of fundamental importance for decarbonizing (or rather defossilizing) the chemical sector, it is equally important to address possible applications of **electricity as an efficient and renewable heating source for chemical processes**. The electrification of chemical processes using renewable electricity is one the most promising pathways towards the transition of low carbon footprint[6].

The electrification of the industry using renewable sources is a step towards reducing the carbon footprint of chemicals. However, this ambitious objective can be only reached if clear targets for chemical processes that could have a large impact at the energy-carbon nexus are identified. The assessment for process electrification consists of first evaluating the options of electrification in the process and heat integration, followed by energy and techno economic analysis that should include not only the energy use but also the greenhouse gas emission assessment[7]. The production of hydrogen as a potential green energy vector could be limited by the substantial CO_2 footprint during the heating of the conventional methane steam reforming process, obtained by burning hydrocarbons. The energy consumption of steam reforming process assisted by microwaves was 3.8 kW/Nm^{3}H$_2$ compared with other electrocatalytic processes ranging from 3.5 to 5 kW/Nm^{3}H$_2$ clearly shows that MW technology deserves a deeper study[8].

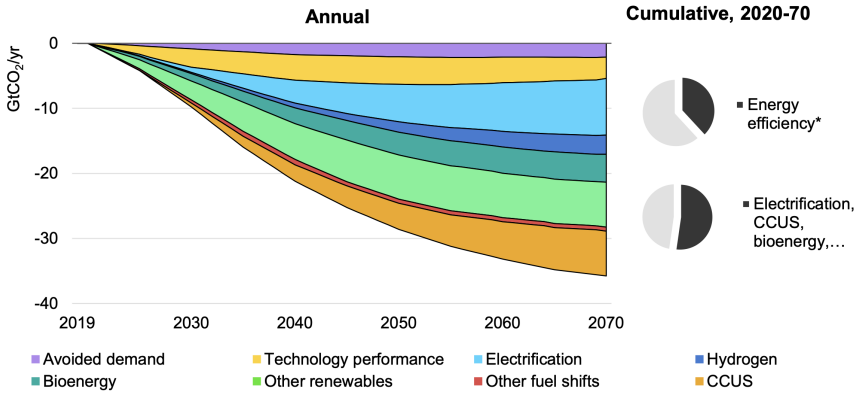


***Figure 1** Reduction of CO2 global emissions for the energy sector in a sustainable development scenario according to sustainable policies in the period 2019-2070.*

The two challenges addressed in RASMIA are the global energy demand and climate change. RASMIA entails the transformation of energy intensive chemical industry through microwave heating MW, a more efficient way of supplying energy to the processes, using renewable electricity avoiding fossil fuels. RASMIA will focus on the design of new **MW heated assisted processes, for CO_2 capture and hydrogen generation by Dry Reforming of Methane, DRM** both aiming to contribute to decarbonization of chemical industry and net zero emissions goal, together with process circularity, since the CO_2 captured in the Microwave Swing Adsorption processs **MWSA** is the raw material for dry methane reforming.

3.2. Justification and expected contribution of the project to the generation of knowledge. Hypotheses.

[5] https://www.odyssee-mure.eu/publications/efficiency-by-sector/industry/energy-consumption-trend-industrial-branch-eu.html Accesed January 2023.

[6] Stankiewicz, A. I.; Nigar, H. Beyond Electrolysis: Old Challenges and New Concepts of Electricity-Driven Chemical Reactors. React. Chem. Eng. 2020, 5 (6), 1005–1016. https://doi.org/10.1039/d0re00116c

[7] Wiertzema, H.; Svensson, E.; Harvey, S. Bottom–Up Assessment Framework for Electrification Options in Energy-Intensive Process Industries. Front. Energy Res. 2020, 8 (August), 1–17. https://doi.org/10.3389/fenrg.2020.00192.

[8] Meloni, E.; Martino, M.; Ricca, A.; Palma, V. Ultracompact Methane Steam Reforming Reactor Based on Microwaves Susceptible Structured Catalysts for Distributed Hydrogen Production. Int. J. Hydrogen Energy 2021, 46 (26), 13729–13747. https://doi.org/10.1016/j.ijhydene.2020.06.299.

MWH is widely exploited in food industry (thawing, tempering, defrosting, cooking, sterilizing and freeze drying)[9]. **Vulcanization** started in the 1960's and is also a well-established microwave process[10]. Material processing also leverages the advantages of the fast and efficient heating afforded by MWs, resulting in energy savings and improved properties of the processed materials including ceramics, metals and alloys, advanced composites and fiber reinforced plastics[11]. However, **looking at the high energy intensive chemical industry there few processes where MWs are implemented**, some examples include polymer processing, waste valorization, and the production of food additives. Recently, Japan based Microwave Chemical has developed a microwave-assisted chemical plant of capacity 10 t/day producing an ink material[12]. This microwave-assisted chemical plant is claimed to reduce the energy consumption, floor area, and reaction time by one-third, one-fifth, and one-tenth, respectively, compared to a conventional plant. Such industrial-scale demonstrations are promising for further commercializing microwave-assisted process intensification techniques. There are **opportunities for developing microwave heated gas-solid contactors, however to achieve the goal of industrial scale it is important to understand the fundamentals and mechanism of microwave heating** and the main challenges that should be faced[13].

The main advantage of microwaves relies on volumetric heating, unlike conventional heating where solids are heated by conduction and convection, microwave field can provide energy to susceptible materials directly, without wasting energy on heating the entire system (e.g. an adsorbent container, assembly, gases, etc.) (**Figure 2**). In this way, the MW field can achieve volumetric heating, i.e. all volume elements within the material are heated in an efficient manner. The ability of a material to be heated in the presence of the MW field is defined by its dielectric loss tangent: tanδ=ε''/ε', where ε'' is the dielectric loss (or imaginary permittivity), which measures the capability of the material to be heated, and it is the dielectric constant (or real part of the permittivity) the one that determines the storage of electrical energy in the dielectric. Furthermore, in the case of the cyclic adsorption/desorption system, **microwave heating** opens up the possibility of achieving **ultra-fast heating rates**, reducing the desorption cycle time with consequent energy savings.

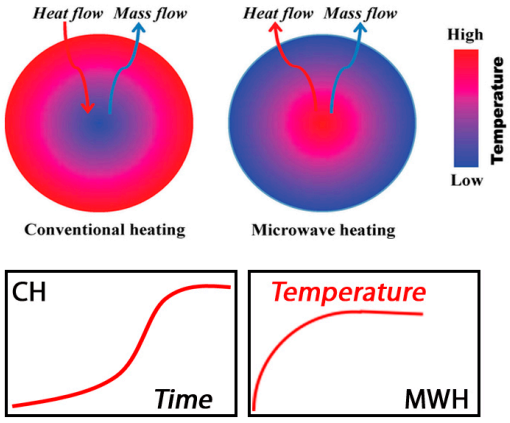


***Figure 2** Microwave heating vs. Conventional heating*

In spite of these advantages and the numerous research papers dealing with the application of MWs to catalytic processes and adsorption in the so-called Microwave Swing Adsorption (MWSA), the technology is not jumping to higher TRL levels because some challenges and difficulties need to be solved. In particular in the case of MWSA, the following issues were identified in a recent review[14]: i) the low penetration of microwaves in high loss adsorbents and ii) the non-uniformity of the electric field strength in solid materials. To overcome these difficulties, several **strategies** are **adopted in RASMIA based on the following hypothesis**: i) Adjustment of the **dielectric loss of the adsorbent materials and structuring of the solids**, resulting in a lower apparent dielectric loss. A compromise must be reached between high losses that increase MW absorption and consequently heating, and the

[9] Microwaves in the Food Processing Industry Academic Press 1985.

[10] B. Krieger, Microwave Vulcanization: A Lesson in Business and Technology, MRS Online Proceedings Library, 347 (1994) 57-69.

[11] D. Agrawal, Latest global developments in microwave materials processing, Mater. Res. Innov., 14 (2010) 3-8.
S. Singh, D. Gupta, V. Jain, A.K. Sharma, Microwave Processing of Materials and Applications in Manufacturing Industries: A Review, Materials and Manufacturing Processes, 30 (2015) 1-29.
R.R. Mishra, A.K. Sharma, Microwave-material interaction phenomena: Heating mechanisms, challenges and opportunities in material processing, Compos. Pt. A-Appl. Sci. Manuf., 81 (2016) 78-97.

[12] https://mwcc.jp/en/factory/m3k.html

[13] Hueso, J. L.; Mallada, R.; Santamaria, J. Gas-Solid Contactors and Catalytic Reactors with Direct Microwave Heating: Current Status and Perspectives. Catal. Today 2022. https://doi.org/10.1016/j.cattod.2022.10.009.

[14] Li, H.; Zhao, Z.; Xiouras, C.; Stefanidis, G. D.; Li, X.; Gao, X. Fundamentals and Applications of Microwave Heating to Chemicals Separation Processes. Renew. Sustain. Energy Rev. 2019, 114 (August), 109316. https://doi.org/10.1016/j.rser.2019.109316

penetration depth that decreases with increasing losses, reducing the volumetric effect. ii) **Proper design and simulation of the MW cavity** considering the dielectric properties of the materials measured as a function of temperature[15].

These two issues also applied for catalytic processes but also, as it is the case of the DRM the operation at higher temperature and production of coke are two important challenges to face. First of all, a reliable temperature measurement and control is needed for MW reactors to achieve thermal stable operation as in conventional heated reactors. Achieving this in MW reactors may be more difficult than in conventionally heated systems, given the strong dependency of dielectric properties with temperature. Thus, in many cases the mobility of charge carriers responsible for microwave heating (MWH) increases with temperature, this causes an instability loop where an increase in temperature increases mobility and microwave absorption leading to further increase in temperature and finally a runaway reaction. Furthermore, in the case of catalytic reactions producing coke, as it is the case in DRM. The key factor for unmanned operation of a catalytic system is based on achieving homogeneous temperature in the system and appropriate measurement, combined with a control loop. All the **MWH systems operating in RASMIA will integrate a temperature control loop**, to ensure stable operation. This control loop is designed by the group at UPV with a wide experience in MWH systems[1617], **as far as we are concerned there is no commercial MW system with this type of control already implemented yet.** Another important aspect to achieve a stable operation is the homogeneous distribution of the catalyst and the heating susceptible materials. The group of UNIZAR has experience on the homogeneous deposition of catalysts by conventional washcoating methods on commercial monoliths that result in homogeneous heating. However, we have observed that these monoliths present sharp edges where carbon deposits occur and they work as an antenna for MWs resulting in hot-spot formation. In **RASMIA we develop structured reactors by additive manufacturing without sharp edges, avoiding hot-spot that would result in runaway reactions**.

Finally, rigorous numerical simulations, as in the case of MWSA, knowing dielectric properties of the materials as a function of temperature and the exact geometry of the structured reactors allow to estimate temperature distributions under MWH, and may be the only tool available for certain systems, where no transparent windows are allowed, i.e. stainless-steel reactor housing for operation at high pressure. The simulations can provide a global picture of reactor temperatures that is beyond the reach of experimental measurement techniques. For this purpose, multiphysics simulation allows coupling electromagnetic field (Maxwell's equations), together with fluid dynamics and heat transfer[18]. In RASMIA we will develop a model for simulation of MWH of structured reactors.

Monoliths and structured solids for adsorption and catalysis

There are two strategies for incorporating the active materials either adsorbents or catalysts into a monolithic structure. A direct way in which the material is incorporated into a paste containing the material, binding agents and solvent and then this paste is directly extruded to form the monolithic structure. Another option is to incorporate the adsorbent in a second stage after extrusion, post-fabrication method, either by i) using immersion methods such as wash-coating of the monolith with a suspension containing the material[19] or by ii) direct synthesis of the material on the monolith wall using a seeding and secondary growth method [20]. In the direct way the main advantage is that it is a

[15] Catalá-Civera, J. M.; Canós, A. J.; Plaza-González, P.; Gutiérrez, J. D.; García-Baños, B.; Peñaranda-Foix, F. L. Dynamic Measurement of Dielectric Properties of Materials at High Temperature during Microwave Heating in a Dual Mode Cylindrical Cavity. IEEE Trans. Microw. Theory Tech. 2015, 63 (9), 2905–2914.

[16] Garcia-Baños, B.; Catalá-Civera, J. M.; Peñaranda-Foix, F. L.; Plaza-González, P.; Llorens-Vallés, G. In Situ Monitoring of Microwave Processing of Materials at High Temperatures through Dielectric Properties Measurement. Materials (Basel). 2016, 9 (5). https://doi.org/10.3390/ma9050349.

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[18] Nigar, H.; Sturm, G. S. J.; Garcia-Baños, B.; Peñaranda-Foix, F. L.; Catalá-Civera, J. M.; Mallada, R.; Stankiewicz, A.; Santamaría, J. Numerical Analysis of Microwave Heating Cavity: Combining Electromagnetic Energy, Heat Transfer and Fluid Dynamics for a NaY Zeolite Fixed-Bed. Appl. Therm. Eng. 2019. https://doi.org/10.1016/j.applthermaleng.2019.03.117.

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[20] Ulla, M. A.; Miro, E.; Mallada, R.; Coronas, J.; Santamaría, J. Preparation of Highly Accessible Mordenite Coatings on Ceramic Monoliths at Loadings Exceeding 50% by Weight. Chem. Commun. 2004, 4 (5). https://doi.org/10.1039/b315386j

less stepwise manufacturing method in which the stability of the adsorbent material in the solvents used and during the extrusion process has to be considered. On the other hand, in the direct method, the sorbent is less accessible compared to post-manufacturing, which offers a higher volumetric loading of active material[21]. Recent advances in **additive manufacturing offer the possibility to incorporate the material directly or indirectly with a customized geometry optimizing the gas-solid contact** (**Figure 3**).

An optimal 3D printing of monoliths requires that the original crystalline structures and textural properties are maintained. In addition, the mechanical, chemical and thermal resistance of the final structure must be suitable for the process. Both direct methods, i.e. direct ink writing (DIW) and post-manufacturing methods including secondary growth and surface functionalization /impregnation are described in the literature. With porous materials, especially inorganic materials, silica and zeolites, direct printing is the most common way. An ink consisting of a suspension containing the materials, binders and a solvent is used. The ink is extruded through a nozzle, and the relative movement between the substrate and the nozzle controls the deposition of the paste. Finally, solvent removal and post-treatment steps such as calcination are carried out to obtain the appropriate mechanical properties.

In the case of MOF-type materials, ink stability is generally not adequate and for this reason there are more examples of seeding and secondary growth and impregnation. In one of the first cases of monolith fabrication based on 3D printing, Thakkar et al. highlighted the possibility of 3D printing monoliths made of aminated silica[22], zeolites or MOFs for CO_2 capture applications[23]. **While there are several examples of 3D printing of adsorbent materials, recently compiled in a review[24], research on the dynamic properties of these monoliths in cyclic processes is scarce**. At RASMIA we investigate both adsorption and desorption processes, using an innovative microwave regeneration method, and performing several cycles to study the evolution of the structured adsorbent and its properties

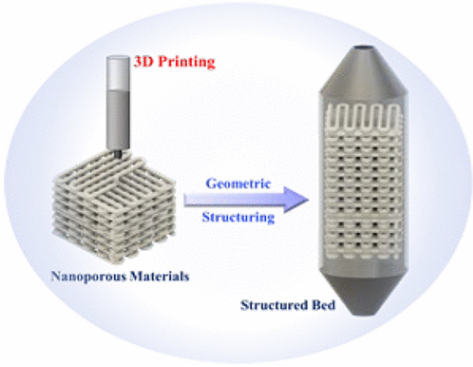


Figure 3 *3D printing concept for structured beds from ref.24*

In catalytic applications it is important the mechanical resistance and thermal conductivity of the printed monoliths specially when dealing with high temperature reactions as it is the case of DRM. The group at "Instituto de Ceramica y Vidrio" worked in robocasting technique for the preparation of monoliths acting as catalytic support. They prepared composite porous structures based on γ-Al_2O_3, and also containing graphene nanoplates GNP to increase thermal conductivity. These GNP may act also as MW susceptors[25].

CO_2 capture

Sorbent materials for CO_2 capture

The alternative of adsorption process vs. the conventional amine adsorption process is based on the potential energy savings in the regeneration process. According to Gray et al 2009[26] , **an adsorbent**

[21] Stephen, D.; Sinha, A.; Kalyanaraman, J.; Zhang, F.; Realff, M. J.; Lively, R. P. Critical Comparison of Structured Contactors for Adsorption-Based Gas Separations. Annu. Rev. Chem. Biomol. Eng. 2018, 9 (March), 129–152. https://doi.org/10.1146/annurev-chembioeng-060817-084120

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[25] Moreno-Sanabria, L.; Ramírez, C.; Osendi, M. I.; Belmonte, M.; Miranzo, P. Enhanced Thermal and Mechanical Properties of 3D Printed Highly Porous Structures Based on γ-Al2O3 by Adding Graphene Nanoplatelets. Adv. Mater. Technol. 2022, 7 (9). https://doi.org/10.1002/admt.202101455.

[26] Gray, M. L.; Hoffman, J. S.; Hreha, D. C.; Fauth, D. J.; Hedges, S. W.; Champagne, K. J.; Pennline, H. W. Parametric Study of Solid Amine Sorbents for the Capture of Carbon Dioxide. Energy and Fuels 2009, 23 (10), 4840–4844. https://doi.org/10.1021/ef9001204

with a capacity close to 3 mmol CO_2/g could significantly reduce the energy requirement by 30-50% compared to amine solvent systems. Since flue gas contains water that can compete with CO_2, this must be considered when selecting an adsorbent, that should have high selectivity towards CO_2 in humid conditions. Other important aspects for adsorbent selection include:

1. High CO_2 adsorption capacity even at low partial pressures. The CO_2 concentration in the flue gas is around 15% and the direct air capture process with concentration of CO2 of around 450 ppm. According to IEA in the Net Zero Emissions by 2050 Scenario, direct air capture is scaled up to capture almost 60 Mt CO_2/year by 2030 [27]
2. High adsorption kinetics. In the case of ultramicroporous materials, the diffusivity of CO_2 is lower.
3. Mild regeneration conditions with minimal energy requirements. Adsorption heats in solid adsorbents vary between 30 and 50 kJ/mol CO_2.
4. The adsorbent must be durable and stable in the operating range, temperatures between 30 and 140ºC, and maintain its structural stability during multiple adsorption and desorption cycles.
5. Tolerance of the adsorbent to other flue gas components, including acid gases and sulphur compounds.

The adsorbents studied for CO_2 capture include microporous zeolites, carbons that tend to have lower capacities, mesoporous amine-modified silicas and MOF (Metal Organic Framework) metal-organic frameworks consist of metal nodes that coordinate by linking to each other through organic linkers, creating three-dimensional crystalline structures. These structures have enormous potential due to their high specific surface area and pore volume, together with their ease of functionalization, thanks to the organic linkers.

Water affects adsorbents differently, with carbons generally being the least affected due to their higher hydrophobicity. The capacity of zeolites in the presence of water, especially those with a lower Si/Al ratio, decreases due to competitive adsorption. In contrast, in the case of amine-modified silica materials[28] the presence of water in the flue gas can be beneficial, increasing their adsorption capacity. These materials show a high adsorption capacity at low CO_2 partial pressures (3.5 mmol/g P_{CO2} = 0.15 atm [29]).

In the case of MOFs, not all structures are feasible for CO_2 capture under wet conditions. The presence of water not only leads to competitive adsorption, but also causes hydrolysis of the metal-linker bond resulting in destruction/alteration of the crystalline structure. However, in some cases the water retained in the structure affects positively increasing the total adsorption capacity up to 5 times as in the case of MIL-101 (Fe)[30]. Due to its high thermal and chemical stability, the MOF material UiO-66 containing a Zr node has been extensively studied in the literature for CO_2 adsorption. To increase its capacity and selectivity, multiple modifications have been performed, including the substitution of part of the Zr by Ti[31], ionization with alkaline solutions[32] , or other functional groups including amino among others[33] , and the use of linkers such as 4-aminobenzoic acid (PABA) as an amino-defective linker[34]. The effect of defects on the structure and adsorption capacity[35] and the preparation of composite

[27] https://www.iea.org/reports/direct-air-capture Accesed January 2023.

[28] Nigar, H.; Garcia-Baños, B.; Peñaranda-Foix, F. L.; Catalá-Civera, J. M.; Mallada, R.; Santamaría, J. Amine-Functionalized Mesoporous Silica: A Material Capable of CO2 Adsorption and Fast Regeneration by Microwave Heating. AIChE J. 2016, 62 (2). https://doi.org/10.1002/aic.15118

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materials such as TiO_2 plus UiO-66-NH_2 [36] and graphene oxide-UiO66 [37] have also been studied. Another of the most studied MOF-type structures for CO_2 capture is Mg-MOF-74, despite the competitive adsorption of water in the structure, very high CO_2 loadings are achieved under conditions of humidity and 15% CO_2 (5.2 mmol/g) [38]. Recently, modifications of this material with various functional groups have been compiled, similar to the case of UiO-66, to increase its capacity and selectivity[39].

These modifications in MOFs in which polar amino groups or molecules with benzene rings containing delocalised electrons are added result in the modification of the dielectric properties of the material, increasing its dielectric constant and therefore its ability to be heated by microwaves. In fact, in a recent work Muschi et al. present GO-MIL-91 composite materials for CO_2 adsorption that is regenerated with microwaves and in which they observe an increase of the dielectric constant of the composite material. [40]

The **innovative aspects in RASMIA** are aimed at the synthesis of silica and MOF-type materials functionalized with amino groups and the preparation of MOF-MWs receptor composite materials that exhibit **high capacity and selectivity for CO_2 adsorption** under relevant conditions (presence of water and 15% CO_2) **and at the same time are capable of efficient, low-energy heating with MWs radiation.**

Structured adsorbents for CO_2 capture

The selection of a suitable gas-solid contact system is key to extract the maximum potential of an adsorbent solid and to increase the CO_2 capture rate[41]. In this sense, structured adsorbents and monoliths offer a great opportunity, due to their low pressure drop and the high volumetric gas flow that can be processed[42]. In the case of amine adsorption, a maximum recovery of 230 tonnes of CO_2/h could be achieved in an adsorption column (with a maximum size of 12.2 m long and the same diameter)[43]. Krishnamurthy recently observed 2.35 times improvement in productivity in a Vacuuum Swing Adsorption (VSA) process performed with a 3D printed sorbent compared with its counterpart using pellets[44]. The potential of structured monolith technology for CO_2 capture has been confirmed by several recent contributions [45],[46],[47],[48]. In all cases, high gas velocities are present. For example, Verougstraete et al.[44] report values ranging from 0.2 to 3.4m/s with contact times between 0.8 and

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0.04s. In this case, RTSA (Rapid Thermal Swing Adsorption) regeneration cycles are performed with steam and lead to higher energy costs compared to VSA regeneration[45]. In this system, a composite adsorbent, also based on carbon, with corrugated channels was used and showed a high performance in the separation with CO_2 streams with a product purity of 40.08% and a recovery of 96.55% CO_2, with a productivity of 2.40 mol CO_2/kgads h, with an energy consumption of 414.86 kJ/kgCO_2 , 30% less than that of the regeneration with amines of 612 kJ/kg CO_2[42].

High flow rates in structured adsorbents, coupled with short adsorption/desorption cycles, are the perfect combination to achieve higher productivity and a lower carbon footprint of the process. In fact, this is the combination of the successful process intensification case developed by Exxon Mobile in the RapAdsorb™ process which is an adsorption-based separation technology for natural gas processing and treatment. RapAdsorb™ uses rapid cycles of pressure and/or temperature variation, in the order of seconds to a few minutes, minimising energy use and drastically reducing the size and weight of the equipment, and the overall carbon footprint of gas treatment by 50% compared to conventional technologies [49].

MWSA for CO_2 capture

The research on CO_2 capture, together with MW regeneration is scarce and limited to solids in powder form or granules with carbon and zeolites,[50],[51], [52],[53,54] and our research, UNIZAR and UPV previous collaboration (see details in section 4.5), on mesoporous silica modified with amino groups being the most cited [27]. In all the cases the results presented include fast desorption rates, high cyclic stability and potential energy savings compared to conventional heating. and for CO_2 capture, in collaboration with UPV, using as adsorbent mesoporous silica, MCM-48, functionalized with amine groups, either with one (mono), two (di-) or three (tri-) amino groups[55]. **Figure 4a)** shows that as the amination increases, the CO_2 adsorption capacity increases, up to values of 70 gCO_2/kg (1.6 mmol/g). At the same time, by including polarizable groups in the silica, the loss tangent increases, resulting in higher efficiency in microwave heating. A higher temperature recorded with the thermographic camera for the di- and tri-samples can be observed in the inset of a). Finally, the adsorption capacity remained practically unchanged after 20 adsorption/desorption cycles. Desorption is three times faster with MW heating compared to conventional heating **Figure 4b)**. **Figure 4c)** shows the experimental system consisting of a TE10 single mode microwave cavity, fed by a solid-state generator and in which a standing wave is generated, whose maximum is adjusted with the help of a stub tuner to the central part where the adsorbent fixed bed is located. Furthermore, in collaboration with UPV and Delft Technical University we simulated the MW heated fixed bed. The heat transfer equations, fluid dynamics and Maxwell Stefan equations for the electromagnetic field are solved simultaneously with the software COMSOL® Multiphysics. **Figure 4d)** shows the simulation of the temperature evolution during microwave heating of a fixed zeolite bed together with the experimental results and shows a perfect temporal and spatial agreement[18].

RASMIA proposes high productivity for CO_2 adsorption in structured sorbents, together with fast temperature regeneration cycles with low energy consumption thanks to rapid microwave heating. As far as we concern, **there are no reports in the literature on structured sorbents for CO_2 MWSA.**

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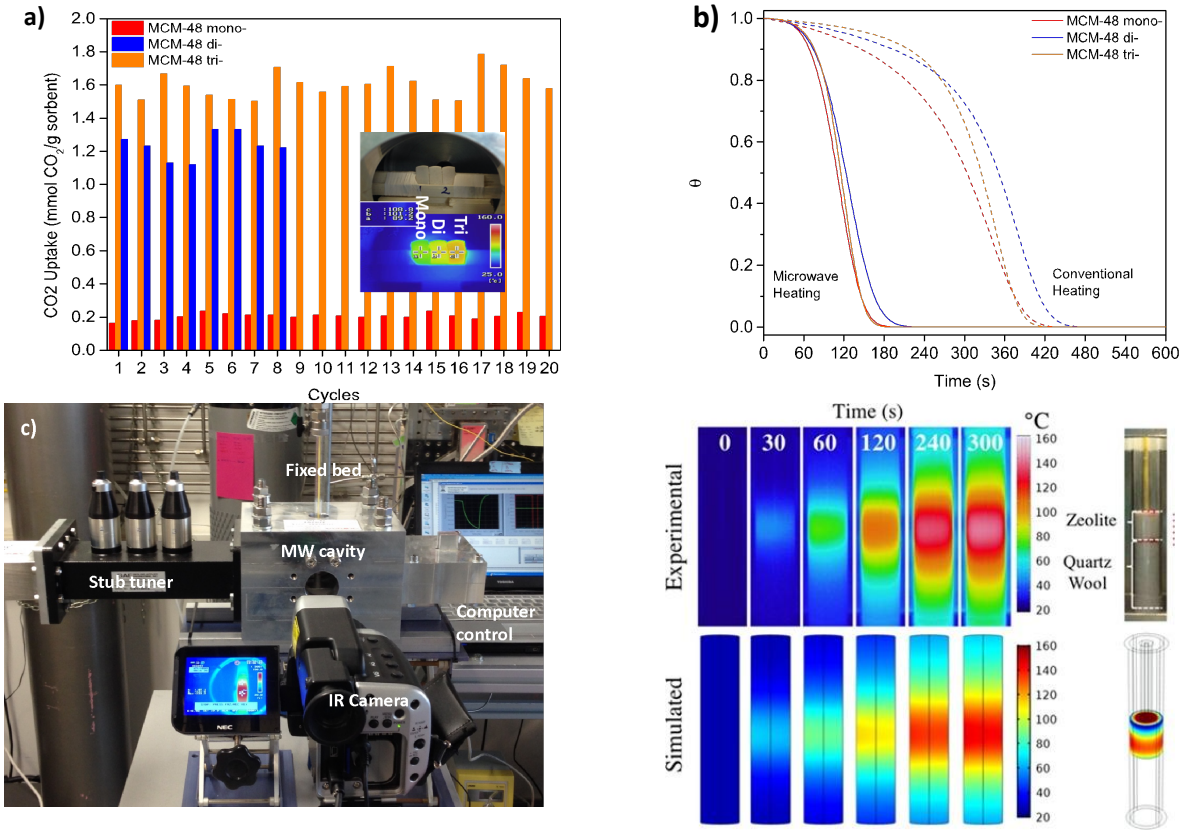


Figure 4 *CO_2 adsorption and regeneration cycles with microwave heating and using N_2 as purge gas. MCM-48 mesoporous silica pellets functionalised with mono- di- and tri- amines heated with microwaves and temperature observed with thermographic camera. b) Occupancy versus time during CO_2 desorption for conventional and microwave heating c) Experimental single-mode microwave cavity system for regeneration of adsorbent beds d) Temperature evolution during heating of a fixed zeolite bed, experimental and COMSOL Multiphysics simulation.*

Reforming of Methane for syngas production

Syngas production

Syngas is the mixture of H_2 and CO, the gaseous feed for the production of the bulk chemicals' ammonia and methanol, the synthesis of hydrocarbons and fuels in the Fischer-Tropsch process and last but not least, the actual source of 95% of the H_2 produced. Syngas with a H_2/CO ratio of around 3 is produced in the conventional industrial methane steam reforming process operating at high pressure and temperatures>800ºC introduced in the 1960s. Reforming of methane with CO_2 instead of H_2O, DRM, results in a H_2/CO ratio of 1 and reduces the carbon footprint by the abatement of two of the most abundant greenhouse gases. There are already some industrial plants for CO_2 rich reforming in operation[56] that used Ni/$MgAl_2O_4$ and Ru/$MgAl_2O_4$ catalyst. DRM of methane is an endothermic reaction (ΔH=247,3kJ/mol) and as result requires elevated temperatures to shift the equilibrium. At these temperatures the main problems associated with catalyst are deactivation by coke deposition and sintering. Furthermore, in the case of methane feed containing sulfur poison of the catalyst reduces activity.

Catalysts for DRM

The conventional catalyst for DRM consists of a metallic active phase deposited on a support material. Metallic active phases include noble metals with excellent performance, limited availability and high price, and more affordable transition metals being Ni the most common. The activity of noble metals at 650ºC goes in the order Ru>Rh, Ni>Ir>Pt>Pd and the carbon deposition on the catalyst follows the order Ni>Pd>>Ir>Pt>Ru,Rh, for catalyst supported on $MgAl_2O_4$[57]. The support roles include the dispersion of the metallic nanoparticles and also in basic supports adsorption of CO_2 is favored. Although γ-Al_2O_3 is a conventional support the formation of $NiAl_2O_4$ spinel at high temperature decreases the activity. Basic metal oxide supports include La_2O_3, MgO, CaO and CeO_2. MgO acts also as promoter in reforming reactions

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and the formation of $MgAl_2O_4$ spinel has shown to limit sintering and carbon deposition[58]. Using the same support, particle size also affects coke formation. Most DFT studies have estimated critical size for Ni nanoparticles around 6-7nm and some experimental data suggest 2nm. Bimetallic catalysts show promising results ascribed to i) a cooperative effect of the two metals increasing active sites ii) electronic effect arising from metal-metal contact leading to lower carbon deposition iii) a sacrificial role played by one of the elements, allowing the second metal to remain unpoisoned and available [60]. Finally, with the advancement of nanotechnology many different reported in the last years. This structure includes core-shell, sandwich configuration or nanotubes, however synthesis scale-up manufacturing is challenging.

Microwave heating for DRM

Microwave assisted dry reforming was pioneered studied by the group of Menendez at INCAR using Ni catalyst supporte on carbon as MW heating susceptible material[596061], the oxidation of the carbon support, reduces the catalytic properties considerably. Although some authors postulated a space-charge mechanism, under MWH, forming electron-hole pairs at the surface of carbon which interact with CO_2[62], Fidalgo et al. found similar conversions over activated carbon FY5 + Ni/Al_2O_3 under MWH and Ni/Al_2O_3 under conventional heating[63]. The formation of local hot spots may induce misinterpretation of the results and MW effects, we also observe previously non-uniform heating and formation of hot spots in the DRM with a perovskite catalyst prepared at UNIZAR and tested in a fixed bed MWH at U. Delft[64].

Reactor configurations in DRM

The use of monolithic type structures in endothermic thermocatalytic process is well stablished in the scientific community as a process intensification solution[65] In general metallic monolithic structures, with high thermal conductivity, more or less complex[66], provide a better thermal energy management. The integration of MHW and structured catalyst MW susceptible, for example a monolith made of SiC coated homogeneously with the corresponding catalyst will result also in a homogeneous heat distribution, and at the same time the temperature gap between gas and solid may result in an increase of selectivity, as we already proof for different reactions[6768]. Meloni et al. recently reported MW assisted heating for methane steam reforming MSR reaction in a NiW-SiC monolithic reactor, with an energy consumption of 2.5kW/Nm^3H_2 very close to the best resistive MSR[69].

In **RASMIA** we propose **SiC structured monoliths as high susceptible material for MW heating, and catalyst support** using different configurations for catalyst deposition and homogeneous heating thanks to **uniform deposition and appropriate design of MW cavity together with a temperature control loop.**

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3.3. Justification and expected contribution of the proposal to solve the problems of the priority research line

RASMIA focuses on two global thematic priorities, climate change and energy demand increase. To contribute to these areas and the reduction of emissions achieving the goal of net zero global CO_2 emissions by 2030, established by EU. RASMIA, in a multidisciplinary team aims to develop a technology based on microwaves for providing heat to chemical processes. In **Table 1** we summarized the contribution of this proposal to solve the specific needs of the technology, based on the analysis of the state of the art.

Table 1 Summary of RASMIA contribution to specific needs and challenges

State of the art	RASMIA beyond state of the art
Microwave heating in chemical processes	
MW technology offers multiple advantages for the realization of different chemical processes, and in particular gas-solid contactors, as it allows the introduction of heat energy directly into the material resulting in higher heating rates, without heating the surrounding parts of the reactor and without contact and electrode-free. MW energy using renewable electricity contributes to decarbonization of the energy-intensive chemical industry. MW heating already proof energy savings in laboratory scale, however to reach higher TRL levels two important aspects need to be solved: i) low penetration of microwaves in high loss adsorbents and ii) the non-uniformity of the electric field strength in solid materials, resulting in hot-spots and possible runaway reaction.	RASMIA develops structured solid-gas contactors with a lower apparent dielectric loss for higher penetration and also ad-hoc materials tailoring dielectric and sorption/catalytic properties, but keeping the value high enough to guarantee the microwave heating up to the requested temperatures. The specific design of the MW cavity, taking into account measured dielectric properties as a function of temperature, together with a temperature control loop ensures the stable operation required in chemical processes. The results of the project will establish the strategies and rules for designing microwaved heated assisted chemical processes. This will be possible thanks to the multidisciplinary team in the coordinated project that brings together expertise in materials science and chemical engineering at UNIZAR with experts in the field of designing MW equipments and instrumentation.
Monoliths and structured solids for adsorption and catalysis and MWH	
Structured reactors and adsorbents in the market consists of extruded monoliths with defined honeycomb structure and made of a limited number of materials	RASMIA develops 3D printed structures, tailor made, based on fluid dynamics simulations to improve turbulence and S-G contact. The structures may include composite materiales giving additional properties such as adsorption, catalysis and at the same time including MW susceptor materials
CO_2 capture process	
The state of the art for CO_2 capture is absorption with liquid amines. The Achilles heel of this process is the use and regeneration of the solvent. Adsorption systems with solids are presented as an alternative, they have higher reaction speeds, lower volatility and specific heat of the adsorbent compared to the aqueous solvent, which means that less energy is consumed during regeneration[33].	Development of MOF-type materials (UiO-66 or MOF-808 and Mg-MOF74) and silica based materials with high CO_2 adsorption capacity and selectivity thanks to the functionalization with amino groups. The synthesis of MOF-type materials will be carried out in an environmentally friendly way. In RASMIA, water or "green solvents" are used as solvents.

Currently the best-known adsorbents are based on carbons and zeolites. In general, carbons have low adsorption capacities and zeolites have low selectivity in humid streams as combustion gases.	The thermal and chemical stability of the sorbents is also pursued under the regeneration conditions.
MWSA for CO_2 capture	
The first literature reports on MWSA adsorption process. Since then the studies focus on zeolites and carbon in water and volatile organic compounds adsorption. So far, research on CO_2 adsorption is limited also to zeolites and carbon in powder or granular form. The results presented include fast desorption rates, high cyclic stability and potential energy savings compared to conventional heating.	In RASMIA we propose the development of completely novel, tailor-made hybrid and/or functionalised structured adsorbents optimised for the CO_2 capture and regeneration process with MWs. These materials include MOFs and aminated silicas. We aim to couple the adsorbents with appropriate susceptor materials to optimize the MWs-adsorbent interaction and thus improve the direct and homogeneous volumetric heating and performance of the selected adsorbents in the MWSA process.
Microwave-driven dry reforming of methane	
DRM represents an excellent alternative to tackle the current levels of two of the greenhouse gases currently contributing to global warming. CH_4 and CO_2 can be simultaneously upgraded into synthesis gas ($CO+ H_2$) with a proper ratio for direct use in multiple industrial processes. The use of MW reactors to perform DRM has been successfully reported in the past years and multiple authors have claimed that MW are able to reduce reaction temperatures (typically above 850 ºC under conventional heating) and partially prevent the formation of carbon deposits, due to their strong interaction with MWs. The translation of MW reactor into a scale-up level still requires further	RASMIA aims at exploring and designing a novel MW reactor with the aid of robust combination of experimental and simulation work in order to correlate the properties of the catalysts under MW operating conditions. In addition, the selection of different packing strategies and different configurations of structured monoliths will be key to maximize the advantages of MW induced heating against conventional heating. Moreover, the influence of different parameters such as the co-deposition of CO_2 adsorbers, co-catalytic phases or deposition methods for the catalytic sites will be key for the advancement of the current state of the art.

4. OBJECTIVES, METHODOLOGY AND WORK PLAN

4.1. General and Specific Objectives

The main goal of RASMIA is the development of MW assisted heating for chemical processes, which results in energy reduction consumption in the chemical industry, contributing to the net zero global CO_2 emissions by 2050.

To achieve this ambitious goal, the specific objectives together with Key Performance Indicators (KPI) in order to evaluate them are detailed below. The work package where the objective is developed and the responsible researcher (RR) of the corresponding subproject is indicated as well.

OBJ1 Development and characterization of functionalized solid adsorbent materials and/or selective compounds for CO_2 capture, capable of being heated with microwaves. (WP1) Subproject 1: RR:RM

KPIs: *Capacity and selectivity:* Adsorbent with a loading capacity of 2 mol CO_2/kg in an effluent with 15%CO_2 and 100% relative humidity. *Cyclability and stability*: Maintains adsorption capacity > 90% of initial capacity after 10 cycles. *MWs heating*: A fixed bed of 200mg solid adsorbent should reach a temperature of at least 120ºC in 3 minutes with an adsorbed MWs power of less than 10W.

OBJ2: Development of a structured system using additive manufacturing with optimal S-G contact and which can be heated with microwaves with low energy consumption. (WP2) Subproject 1: RR:JLH

KPIs: Structured system, with dimensions of 1.5cm diameter and 3cm height, with low pressure drop and optimum G-S contact, mechanically stable and resistant to heating cycles in MWs, at least 20 cycles, in the operating temperature range of the process.

OBJ3: Development of a MWSA process obtaining a concentrated CO2 stream from a gaseous effluent of simulated postcombustion gases. (WP3) Subproject 1: RR:RM

KPI: A laboratory-scale cyclic adsorption system with heating and cooling cycles of less than 4 minutes duration (each), in which at least 250 kg CO_2/h.m^3 adsorbent can be processed to produce a desorbed CO_2 stream with a purity of at least 50 %.

OBJ4: Development of a structured catalytic system for the methane dry reforming reaction, which operates stably under microwave heating. (WP4) Subproject 2: RR: FPF

KPIs: *Conversion*: Stable opearation in the system without formation of hot spots and achieving conversions greater than 90% for methane and CO_2 for a space velocity of 5000 h^{-1}, with a catalyst deactivation lower than 10% after 12h of operation. *Energy efficiency*: Energy consumption must be less than 4 kW/Nm^{3}H$_2$. *Stability of heating MWs*: The System shall opérate stable with unmaned with operation thanks to control loop able to modify EM energy deposition as a function of the evolution of the dielectric properties of the catalyst, including coke formation.

OBJ5: Scale up of a cavity heated by microwaves for a chemical process unit. (WP5) Subproject 2: RR: FPF

KPIs: Design and construction of an MW applicator working at 915 MHz that processes simples at least 150 times volume, the samples processed with the selected process (the fulfilment of KPIs associate to OBJ3 and OBJ4 will be used to decide the optimum process). The system should operate continuously in unmanned operation for at least 100h in the case of continuous DRM and in case of MWSA the system should operate stable for at least 30 cycles.

4.2. Methodology description

In RASMIA we develop specific methodologies to achieve efficient MWH for chemical processes. These include i) specific design of materials considering not only the adsorptive or catalytic properties but also the interaction with microwaves ii) study of dielectric properties as a function of temperarure with exclusive equipment developed at UPV iii) incorporation of the materials in 3D structures prepared by different methods including additive manufacturing iv) microwave applicator specifically manufacture for the chemical processesin a multidisciplinary team, with experts in design and manufacturing these type of equipment based on simulations considering dielectric properties. Details about the methodology are described below in the workplan, together with the viability. The associated risks and contingency plan are described in **Table 3**.

4.3. Work plan and Chronogram

The research team in Subproject 1 includes 5 researchers, 4 of them belonging to the University of Zaragoza, including 3 researchers from the Nanostructured Films and Particles NFP group and 1 researcher from the Manufacturing Engineering and Advanced Metrology Group GIFMA and there is also a researcher from the Miguel Hernández University. The expertise of each of them is described below.

Prof. Reyes Mallada (RM), is full professor of Chemical Engineering at the University of Zaragoza, with 25 years of research experience in the area of **porous and structured materials and metallic nanoparticles** for applications in **catalysis,** membranes, **adsorption** and chemical sensors. In 2012 she started researching in the area of **microwave heating for applications in chemical processes**. Since then she has co-authored 23 publications in this field, co-supervised 3PhDs and participated in 3 EU projects developed in the NFP group on this subject, in which the other co-IP of this project José Luis Hueso, has also participated.

Dr. Jose Luis Hueso (JLH), is a senior researcher with a wide experience in the field of **materials and catalysis** and since 2011 he is a member of the NFP group at UNIZAR. Current research activities are related with the design of efficient catalysts susceptible to microwave, laser and/or light activation as well as the development of novel reactor configurations activated by electromagnetic radiation. He has more than 10 years' experience in microwave heating for chemical processes.

Dr. Pilar Lobera (PL) is profesor contratado doctor at UNIZAR and her research interest include **catalysis**, with a wide experience including a post-doctoral stay in "Instituto de Tecnología Química in Valencia, nanosafety and **development of nanomaterials**.
Dr. Lucia Diaz (LD) received her PhD in industrial engineering in 2019 and since 2020 is Profesor Ayudante Doctor, (accredited for Profesor Contratado Doctor since 2020). Her research career has been developed within the research group, GIFMA which focuses on precision engineering and dimensional metrology applied both to the design of new equipment and to the improvement of the performance of existing equipment such as **computed tomography, additive manufacturing equipment** and optical measuring microscopes.
Dr. Felipe Hornos (FH) is Chemist and received his PhD in November 2020, in Molecular and Cellular Biology from Universidad Miguel Hernandez where he is associate professor. In November 2021 he joined the NFP group with a postdoctoral fellowship APOSTD from Generalitat Valenciana. He has worked on the **synthesis, functionalization and characterization of nanomaterials**, loaded which different enzymes, in order to be used as **catalysts** in the tumor microenvironment.

The researchers of Subproject 2 include 3 researchers in the research team and another one in the work team, below is detailed their background and expertise in the field of microwaves.
Prof. Felipe L. Peñaranda Foix (FPF) is a Full Professor at the UPV and Deputy Director of International Relations at the UPV School of Telecommunications Engineering. He is also co-leader of the Microwave Division (DIMAS) of the ITACA Institute of the UPV. He has more than 25 years of experience in the fields of teaching and basic and applied research, having developed his research work always in the field of microwave engineering. He has participated in more than 100 research projects, with both private and public funding, and all in the field of **microwave Engineering**. He has published more than 150 journal articles and conferences, has directed 4 theses and has 4 patents.
Prof. Jose M. Catalá Civera (JMCC) is a Full Professor at the UPV and Director of the University Research Institute ITACA. **Co-leader of the Microwave Area of the ITACA Institute**, he has more than 25 years of experience in scientific and applied research initiatives, development and technology transfer in the field of microwave engineering. He has participated in more than 130 research projects with both public and private funding, with more than 200 papers published in scientific journals and conferences and 26 patents, most of them with international extension.
Mr. Antoni J. Canós Marín (ACM) is Collaborating Professor at the UPV and researcher at the ITACA Institute. Member of the Microwave Area of the Institute, he has more than 20 years of experience in scientific and applied research initiatives, **development and technology transfer in the field of microwave engineering**. He has participated in more than 80 research projects with both public and private funding, with more than 50 papers published in scientific journals and conferences and 6 patents, most of them with international extension.
Finally, **PhD. Pedro J. Plaza González (PPG)** is a Doctor Telecommunication Engineer, that will be participating in the working team, because of his **expertise in constructing microwave devices (applicators, etc), with an experience of more than 20 years**

The Workplan is deployed next, as a series of activities and tasks. At the end of this section a chronogram gives the temporal planning of activities and tasks. The researchers involved in each task are given according to the code of their initials presented above and also **ESR1** and **PDP** which are the Early Stage Researchers in training (FPI in the two subprojects). **PD** is the postdoc hired in subproject 1 and **ENG** is the engineer hired in subproject 2. **Figure 5** shows the relationship among the different work packages in this project together with the duration and the responsible researcher for each one.

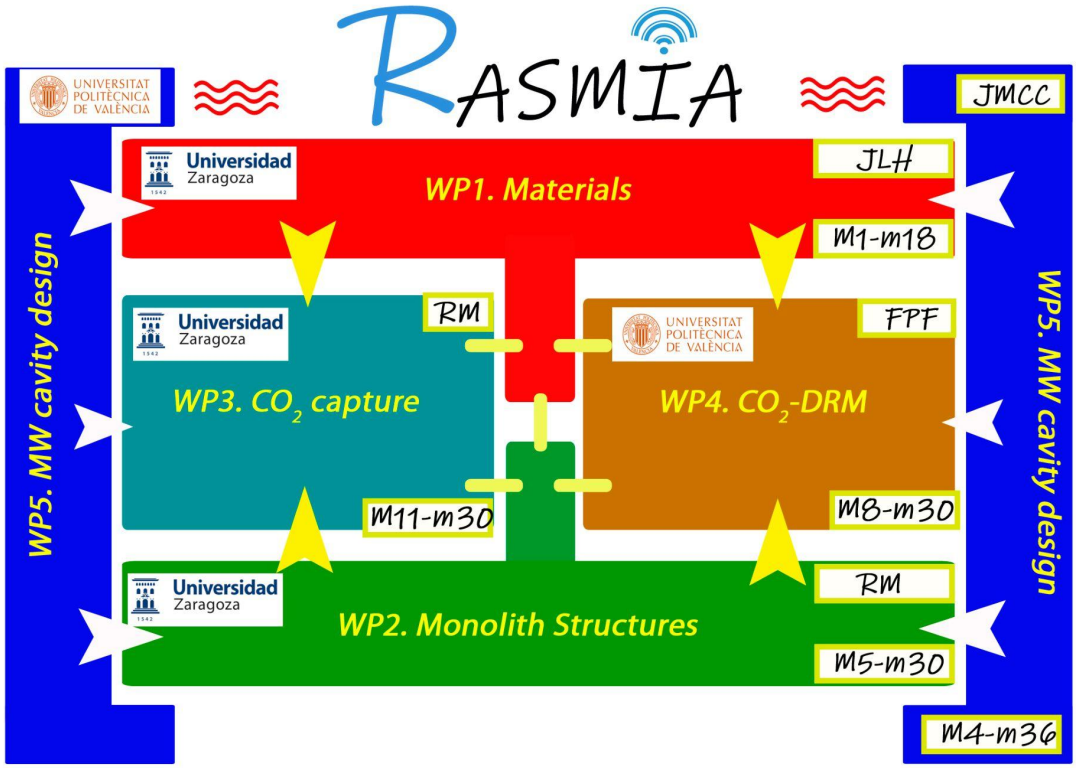


Figure 5 *Relationship among the different WPs in RASMIA, together with group leaders for each WP, identification of leading subproject for each WP (identified with University logos) and expected execution times*

WP1: M1-M18 Material synthesis and characterization (UNIZAR, SubP1-JLH)

In this WP UNIZAR will be responsible for the synthesis of pure materials or composites that could be heated under MWs and at the same time possesses adsorptive and catalytic properties. The dielectric properties of the materials will be evaluated at UPV.

T1.1 Synthesis and characterization of porous materials for CO_2 capture (UNIZAR-**FH**, RM, ESR1)

Synthesis of MOF materials based on Zr (UiO-66 and MOF-808) and Mg-MOF74. Whenever possible, water will be used as solvent or "green solvents" and the possibility of scaling up the process will be considered. Washing and activation of the material after synthesis. Evaluation of the synthesis yield. Functionalization with amino groups, post-synthesis with ligand exchange or in-situ during the synthesis process.

Amination of commercial precipitated silica and mesoporous silica materials by silanisation with tri-amino-silane compound, 2-[2-[2-(3-trimethoxysilylpropylamino)ethylamino]ethylamine and impregnation with polymers polyethylenimine (PEI), polyallylamine (PAA) and polyaniline (PA).

Preparation of composite materials containing MW susceptors. Incorporation of materials with high susceptibility (graphene oxide, carbon nanotubes, semiconductor oxides with high dielectric constants, perovskites...), to the adsorbent solids to improve microwave heating. The loadings will be less than 5% by weight. Preparation of the composites by mechanochemical methods, grinding and crushing. Incorporation during synthesis in "one-pot" processes.

The analysis of CO_2 adsorption isotherm will give information about the maximum CO_2 adsorption capacity for the materials synthesized. The materials with a capacity equal or higher than 2 mol CO_2 /kg adsorbent will be selected for studies in T3.1. Other structural, physicochemical and textural properties characterization techniques include: XRD, Ar/N_2 adsorption isotherms, TGA, Raman, FTIR, electron microscopies SEM-EDX, TEM-STEM.

T1.2 Synthesis of catalytic materials for dry reforming of methane (DRM) (UNIZAR-**JLH**, FH, PL)

Different synthesis strategies following impregnation, sol-gel and/or mechanochemistry will be explored aiming at developing Ni-based catalysts dispersed in a series of hybrid supports containing basic sites to maximize the interaction with CO_2. Alumina, Alumina-Ceria, hybrid spinel and perovskites (e.g. $NiAl_2O_3$, $AlMgO_x$, $MgAl_2O_4$, $CaAl_4O_7$, $SrTiO_3$) NiMg Hydrotalcites will be explored. In addition, the influence of adding active promoters to induce the capture of CO_2 prior to the reaction step will be

explored by the co-addition of CaO, MgO or K_2O and the co-deposition of a second set of transition metals (e.g. Ru, Rh or Co). Alternative catalyst options that will be potentially explored include the MW-susceptible Mo and W carbides or the commercially available Rh/Al_2O_3 benchmark catalyst. The same characterization techniques as in T1.1 together with HRTEM when needed will be employed. Active catalyst prepared for delivery to T2.1 and later on tests on the MW cavity T4.2 will be prepared by M12, however a continuous iteration with T4.1 to improve activity and resistance to deactivation will continue till M18 (see **Table 2**).

T1.3 Microwave heating and study of dielectric properties (UPV-**ACM**, ESR2)

In this task, we will measure the dielectric properties of synthesized materials while applying microwaves at different temperatures, up to the temperatures relevant for the processes studied i.e. around 200 ºC for desorption and 800 ºC for catalytic reactions. The equipment for these measurements has been developed by the researchers in DiMAS group. We will also measure the materials' electric conductivity and how much microwave energy they can absorb at different temperatures. This information is important for designing the microwave tool and the process in future phases of the project.

M1. Synthesis of porous sorbents with adsorption capacity> 2mol CO_2/kg
D1 Report with synthesis protocols for catalytic material synthesis together with characterization of the materials ready for WP4
D2 Report with the results from electromagnetic characterization of selected materials, dielectric properties and microwave energy absorption.
M2. Porous sorbents able to heat up to 120ºC with 10W microwave power.

WP2 M5-M30. Development of monolithic structured materials for microwave heating (UNIZAR-SubP1-RM)

T2.1 Deposition of sorbents and catalytic materials in commercial monoliths and DIW fabricated structures (UNIZAR-**PL**, RM, ESR1)

In the case of materials with low MW absorption or if there are diffusional limitations to access the adsorptive/catalytic material, the deposition of layers of adsorbent material on either commercial monoliths (C or SiC) is considered. The commercial structures will be also useful for starting simulation of MWH in structured materials. For this deposition, the washcoating process and the growth of layers of adsorbent materials by means of seeding and solvothermal synthesis is proposed, based on previous experience of the research team at UNIZAR. The structures or structures developed in T2.4 will be also coated with the most successful techniques for each material.

T2.2 Microwave heating of structured materials and simulation of electric field (UPV-**FPF**, ESR2)

Once the dielectric properties are known, and in parallel with the cavity design (WP5), a model of the electromagnetic behavior of the samples will be done with the permittivity measurements. The aim of this task is to model the electromagnetic field inside the designed structure, to evaluate the expected field distribution under MW energy. The potential change of the dielectric properties with the temperature (measured in T1.3) will be considered in this model.

T2.3 Design and fluid-dynamic study of structures (UNIZAR-**RM**, LD, ESR1)

Computational fluid dynamics (COMSOL Multiphysics software) is used to study the hydrodynamic behavior of the structures and transport phenomena, with the aim of improving the gas-solid contact, increasing the speed of gas and turbulence through the structured adsorbent, while keeping pressure loss in the permissible level. Different woodpile structures, the most common in DIW, will be simulated using as parameters the diameter of the extruded filaments, their slope and distance between them. Commercial monolith structures will also be simulated to compare them with the ones designed. The most optimal designs will be converted to .STL files and they will be printed in T2.4.

T2.4 Development of structured adsorbents by means of DIW additive manufacturing (UNIZAR- **LD**, FH, ESR1)

Formulation of the ink containing adsorbent material (up to 85%wt.), prepared in T1.1. mixed with an inorganic binding agent (0-15%wt. bentonite or colloidal silica), and organics (0-5%wt. polymers e.g. PVA), plasticizer agent (up to 3%wt. Glenium 303) and a solvent such as water, or an organic solvent that evaporates faster. The rheological properties (flowing effort, shear stress, pseudoplasticity, viscosity,…) are evaluated for each paste since they determine the printability and the behavior during the printing process. The printer, specifically acquired for this project, will be an open access 3D printer, (resolution in X and Y-axes of 20 mm, resolution in Z-axis lower than 5 mm) with a print head based on an endless piston with a microdispenser. After the printing process it is necessary a hardening/firing step to remove binders and also to give mechanical and thermal stability. Typically the temperature range for this process is 500 to 800ºC for a period of 6h. Also in the frame of this project, if we include MW susceptors in the ink a calcination step under MW heating could be considered. The printing diameter will be 1 cm and height up to 4 cm. The mechanical and textural properties, the morphology, composition and distribution of the components will be studied by tomography and with SEM-EDX analysis.

D4 Report with the electromagnetic field distribution as a function of the temperature in the monolith structures.
D5 Report on the fluid dynamic simulation of structures and selection of the best performing in terms of pressure drop and S-G contact.
M3. Define an ink formulation printable that retains mechanical properties of the monolith after printing and also high specific surface area and pore volume after firing step.
M4. Fabrication of monolithic structures efficiently heated by microwaves.

WP3. M11-M30. CO_2 capture, MWSA (UNIZAR-SubP1-RM)

T3.1 Fixed-bed adsorption kinetic studies and breakthrough curves (UNIZAR-**PL**, RM ESR1)
These tests will be performed for at least four of the most promising adsorbents (i.e maximum adsorption capacity >2 mol CO_2 /kg adsorbent prepared in T1.1. Pelletization of all solids to the same particle size, small enough to avoid internal diffusion problems and at the same time big enough for avoiding high pressure drop. Prior to adsorption, the adsorbent is degassed by heating to 150ºC in a N2 stream. Then, the 15%CO_2/N_2 mixture is passed through a 200 mg fixed bed with RH=100% and the gases at the outlet are analyzed in a mass spectrometry system, obtaining the breakthrough curve. The kinetic parameters of the curve, saturation time t_s and breakthrough time t_b=10%t_s, and total adsorption capacity are evaluated.

T3.2 Regeneration tests microwave-heated structured adsorbents and stability in adsorption/desorption cycles (UNIZAR-**RM**, JLH, ESR1)
The same experimental system, feed and gas analysis, used in T3.1 for solids adsorption, is valid for adsorption/regeneration studies in monoliths prepared in T2.1. and T2.4. The microwave cavity specially designed by the UPV team in T5.1, for heating monolithic systems, with the corresponding dielectric properties of the materials T1.3 and based on simulations in T2.4, will be implemented in the set-up in UNIZAR. The MW source already available is a solid-state generator working in a frequency range 2,47 to 2.54GHz. During the experiments the frequency will be tuned to decrease the reflected MW power and increase the efficiency of MWs at different power and consequently different regeneration temperatures. The regeneration will be performed with low gas purge or vacuum flow rates, to obtain the maximum CO_2 concentration in the output stream.

T3.3 Energy and economic analysis of the MWSA process (UNIZAR-**LD**, RM, ESR1)
Once the structured adsorbents have been developed, the kinetic tests of adsorption in a fixed bed and regeneration of adsorbents heated by microwaves have been carried out and their stability in adsorption/desorption cycles has been analysed, the energy needs of the overall process will be calculated. The aim is to quantify both the energy required for CO_2 capture and compare it with current

technology (chemical adsorption with different types of amines) and the impact on different systems (thermal energy for amines and electrical energy for microwaves). KPIs (Key Performance Indicators) of the system will be proposed and quantified to quantify how relevant variables influence the energy required per unit (kg, ton) of CO_2 separated. These variables include: the power and time required for regeneration, the purity and quantity of CO_2 separated, the operating temperature, the degree of vacuum and the number of cycles. With this information, the conditions required to achieve the economic viability of the system will be established. Techno-economic and greenhouse gases emission assessment will be performed. The goal is calculating the possible reductions in greenhouse gas emissions versus the capital and variable cost compared to the reference case.

D6. Report containing kinetic parameters of breakthrough curves for the selected sorbents with high capacity.
D7. Techno-economic assessment of MWSA process vs amine absorption state of the art technology for CO_2 capture.
M5. MWSA lab scale system, KPI: 4 minutes heating and cooling cycles, processing 250 kg $CO_2/h{\cdot}m^3$ with a purity of the outlet stream of at least 45 % CO_2.

WP4 M8-M30: Dry Reforming of Methane using CO_2 (UPV-SubP2-FPF)

T4.1 Evaluation of catalytic activity under conventional heating (UNIZAR-**JLH**, PL, PD)
The catalytic materials described in T1.2 will be initially tested using an electric oven. Different catalytic beds will be systematically compared, including self-sustained sieved solid, packed catalyst with ground SiC sieved pieces and structured monoliths containing a thin coating of catalyst or completely packed sieved catalysts within the monoliths. Spatial velocities in the range of 1-10 L $gcat^{-1}$ h^{-1}, reaction temperatures between 700-900 ºC and CH_4:CO_2 feeding ratios between 0.8 and 1.0 will be further explored. The potential generation of coke deposits and subsequent clogging will be prevented with the design of quartz reactors with narrowing tube sections and avoiding sharp and uneven structured monoliths. The most active catalysts will be delivered for MW-assisted DRM in T4.2.

T4.2 Evaluation of catalytic activity under MWH (**FPF**, JLH, PD, ENG)
The design of the experiments will need the coordination of both partners, UNIZAR will design the reaction bed for the selected catalyst prepared in T1.2 and UPV will work with the cavity designed in WP5.
Different configurations will be considered for the conformation of the reaction bed since the catalysts will be deposited as: i) coated forming a thin film within a structured SiC monolith; ii) sieved and packed with ground SiC pellets or iii) packed within a structured SiC monolith. The use of SiC pellets or structured SiC monoliths will ensure the MW-induced heating of all the catalysts, regardless of their MW susceptibility and will reduce the probability of generating local hot spots.
The reaction experiments will evaluate the performance of the designed applicators (WP5) for the catalytic DRM. Parameters involved are microwave power absorbed, temperatures, uniformity as well as the designed cells for DRM. UPV ensures a correct design of control loop to avoid the formation of hot spots during reaction.

T4.3 Energy and economic analysis DRM (UNIZAR- **PL**, JLH, RM, PD)
From UPV, the measured values are forward and reflected power, to evaluate the sustainability of the process from the energy point of view. Other parameters are temperature distribution, process time, etc. All these parameters will arise from each experiment, and they will use the devices designed and tested in WP5.

D8. Report on catalytic activity of materials prepared and material selection
D9. Techno-economic assessment, including also CO_2 footprint of DRM process vs conventional steam reforming heated by fuels.
M6. DRM lab scale system, KPI: conversions greater than 90% for methane and CO_2 for a space velocity of 5000 h^{-1}, with a catalyst deactivation lower than 10% after 12h of operation. Energy efficiency: Energy consumption must be less than 4 kW/Nm^3H_2

WP5 M4-M36. Development MW systems for MWH chemical processes (UPV-SubP2-JMCC)

Objectives: Design and manufacture of new benchtop MW applicators and coupling antennas to ensure uniformity and efficiency of the MW radiation applied to the samples. A range of microreactor systems and flow reactor modules will be developed to reach fundamental understanding of microwave heating interactions.

Task 5.1 Design of reactors for lab-scale MW-assisted applicator (UPV-**JMCC**, FPF, ESR2)

The development of a novel MW lab-scale reactor for the study of the considered catalytic reactions will be carried out, studying several parameters like the optimal scale, material integration and the overall device. 3D Electromagnetic simulation tools (QW3D and/or COMSOL) will be used to delimit MW thermal distributions and uniformities in the MW applicators. After determination of materials dielectric properties (WP1) and their evolution during the chemical processes, EM simulations will permit adjusting MW modules dimensions to work at the ISM selected frequencies.

Critical variables such as temperature, gas throughput, power, pressure, etc. will be identified and the best monitoring techniques will be selected in terms of technical features and availability. MW filters will be also designed to prevent any MW leakage and the selection of MW ancillary equipment will complete material processing system

Task 5.2. Manufacturing and validation tests (UPV-**PPG**, FPF, ACM, ESR2, ENG)

In this task the evaluation of different functionalities considering the frequency, power, changes in the samples, heating/cooling cycles, etc. will be performed. This task will also ensure that applicable safety directives are fulfilled (i.e. Directive 2004/40/EC of the European Parliament on the minimum health and safety requirements regarding the exposure of workers to the risks arising from physical agents (electromagnetic fields)).

Task 5.3 Engineered design of MW assisted device (UPV-**PPG**, FPF, ACM, ESR2, ENG)

Design of a reactor for MW applications, considering several parameters like the optimal scale, material integration and the overall device. Appropriate software control routines and/or interfaces, trigger signals for synchronization purposes will be developed and process requirements, automatic control functionalities determined. Critical variables such as temperature, throughput, power, pressure, etc. will be identified and the best monitoring techniques will be selected in terms of technical features and availability. Additional safety parameters, as sensors, data management, supply failures, overpressure, off-gas analysis, etc. will be considered.

Task 5.4 Design of scale-up MW reactor and setup (UPV-**JMCC**, FPF, ENG)

This task starts in M31, after finishing WP3 and WP4, associated to OBJ3(M5) and OBJ4(M6) and their corresponding KPI. At this point if both KPI indicators are achieved, the PIs based on the results obtained and also after consultations with stakeholders (see Tablexx), will decide on the system that must be scaled. In case only one KPI is achieved this will be the selected system for scale up.

In this task, the scaling up of the MW system to an intermediate pilot scale, containing a monolith with 150 times bigger volume, with higher throughput based on the lab-scale results and the optimized layers and materials for MW-driven operation. The scaling-up based on intermediate MW modules will be considered to simplify the validation and optimization of both the microwave reaction chamber and the rest of microwave subsystems.

After determining the material dielectric properties during the reaction, electromagnetic simulators will be employed to adjust the dimensions of microwave modules to work at the industrial frequency of 915 MHz (longer wavelength than that used at the laboratory scale). In addition, microwave filters will be adjusted to the new dimensions in order to avoid any microwave leakage, and the selection of microwave ancillary equipment will be made to complete the system for material processing.

D10. Applicator design, consisting of maps of the proposed cavities
D11. Applicator 1, to be used with the proposed cells #1
D12.- Applicator 2, to be used with the proposed cells #2
D13.-Full report of the designed and delivered cavities, including control software, security issues, etc
D14.- Design and constructed scaled-up applicator
M7. Microwave heated chemical process scale-up system working continuous unmaned operation for at least 100h or 30 cycles

Table 2 *Gantt Chart for RASMIA project*

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
WP1	**Material synthesis and characterization OBJ1 (SubP1-JLH)**												M2																								
T1.1	Synthesis and characterization of porous materials for CO2 capture (**FH**, RM, ESR1)								M1																											
T1.2	Synthesis of catalytic materials for dry reforming of methane DRM (**JLH**, FH, PL)												D1																							
T1.3	Microwave heating and study of dielectric properties (**ACM**, ESR2)												D2																							
WP2	**Development of monolithic structured materials for MWH OBJ2 (SubP1-RM)**																														M4						
T2.1	Deposition of materials in commercial monoliths and DIW fabricated structures (**PL**, RM, ESR1)										D3																									
T2.2	Simulation of microwave heating structured materials (**FPF**,ESR2)												D4																							
T2.3	Design and fluid-dynamic study of structures (**RM**, LD, ESR1)															D5																				
T2.4	Development of structured adsorbents by means of DIW additive manufacturing (**LD**, FH, ESR1)																								M3											
WP3	**CO2 capture, MWSA OBJ3 (SubP1-RM)**																																			
T3.1	Fixed-bed adsorption kinetic studies and breakthrough curves (**PL**, RM, ESR1)																D6																			
T3.2	Regeneration tests microwave-heated structured adsorbents (**RM**, JLH, ESR1)																														M5						
T3.3	Energy and economic analysis MWSA of CO2 (**LD**, RM, ESR1)																														D7						
WP4	**MWH Catalytic Methane Dry Reforming OBJ4 (SubP2-FPF)**																														M6						
T4.1	Evaluation of catalytic activity under conventional heating (JLH, PL, PD)																		D8																	
T4.2	Evaluation of catalytic activity under MWH (**FPF**, JLH, PD, ENG)																																				
T4.3	Energy and economic analysis DRM (**PL**, JLH, RM, PD)																														D9						
WP5	**Development MW systems for MWH chemical processes OBJ5 (SubP2-JMCC)**																																				M7
T5.1	Design of reactors for lab-scale MW-assisted applicator (**JMCC**, FPF, ESR2)															D10																				
T5.2	Manufacturing and validation tests (**PPG**, FPF, ACM, ESR2, ENG)																		D11					D12												
T5.3	Engineered design of MW assisted device (**PPG**, FPF, ACM, ESR2, ENG)																															D13					
T5.4	Design of scale-up MW reactor and setup (**JMCC**, FPF, ENG)																															D14					

4.4. Identification of risks and contingency plan

Table 3 Risks identified in RASMIA and contingency plan

Risk	Contingency Plan
Limited efficiency in CO_2 capture of the structured adsorbent	Adapt and adjust the system design based on the experience of the researchers and the determined properties of the sorbent in WP1.
Non-uniform MW heating	Design of a new cavity based on the dielectric properties of the materials studied in WP1. Use 915MHz frequency to increase penetration. Alternatively, the allocation of different non absorbent dielectric sample to redistribute the electromagnetic field inside the cavity is proposed
Microwave heating of adsorbent materials is not efficient and consumes a lot of energy.	Deposit adsorbent materials on SiC monoliths with high susceptibility to heating MWs.
Very slow adsorbent cooling	Moderate heating or improve cooling with improved adsorbent design, increased surface area
Faster deactivation of Ni catalyst under MWH compared to conventional heating	Development of alternative catalysts containing new formulations with more basic sites or co-catalysts able to reduce the formation of carbon deposits
The messages of the project do not reach the different audiences, scientists, industry and society	All possible communication channels will be carefully considered so that relevant stakeholders are targeted.

4.5. Preliminary results and expertise from the participants in correlation with the proposal

First of all, it is important to highlight that the groups of the two projects NFP and DIMAS, in a previous collaboration, developed amine-functionalised mesoporous silica materials for CO_2 adsorption, regenerated with microwaves (see results in **Figure 4**).

In the NFP research group at UNIZAR, we have been working on microwave heating technology for gas-solid contactors since 2016. The "Microwave Swing Adsorption" MWSA concept was developed in the framework of the HECTOR project[70], for two chemical processes, the desorption of volatile organic compounds VOCs and subsequent catalytic destruction[71], and for CO_2 capture[28]. In the field of solid-gas catalytic systems UNIZAR studied the ethylene epoxidation in structured reactors developing a methodology for the measurement of solid and gas temperature with the main finding that a solid-gas temperature gradient exists under microwave heating[72]. This finding was explored to improve selectivity in dehydrogenation reactions favoring selective reactions in the catalytic surface versus non-selective reactions occurring in the gas phase.[67]

Finally, the EU Project ADREM[73] "Adaptable Reactors for Resource- and Energy-Efficient Methane Valorisation" the methane dehydroaromatization was studied in a monomode cavity designed by UPV

[70] HECTOR: Microwave-Assisted Reactors: Development of a highly efficient gas phase contactor with direct catalyst heating. ERC Advanced Grant. GA 267626 IP. Jesús Santamaría. Periodo 1/03/11 a 28/02/17.

[71] Nigar, H.; Navascués, N.; De La Iglesia, O.; Mallada, R.; Santamaría, J. Removal of VOCs at Trace Concentration Levels from Humid Air by Microwave Swing Adsorption, Kinetics and Proper Sorbent Selection. Sep. Purif. Technol. 2015, 151. https://doi.org/10.1016/j.seppur.2015.07.019

[72] Ramirez, A.; Hueso, J. L.; Mallada, R.; Santamaria, J. In Situ Temperature Measurements in Microwave-Heated Gas-Solid Catalytic Systems. Detection of Hot Spots and Solid-Fluid Temperature Gradients in the Ethylene Epoxidation Reaction. Chem. Eng. J. 2017, 316. https://doi.org/10.1016/j.cej.2017.01.077

[73] ADREM: Microwave-Assisted Reactors: Development of a highly efficient gas phase contactor with direct catalyst heating. ERC Advanced Grant. GA 267626 IP. Jesús Santamaría. Periodo 1/03/11 a 28/02/17.

with an automatic PID control that enables to follow established heating ramp to achieve reaction temperatures as high as 900ºC, and control under operating conditions while coke was produced[74]. The catalytic system consists of a SiC structured monolith wash-coated with Mo-ZSM5 catalyst. The system was scale-up and tested in Technical University of Denmark – DTU[68].

There is also experience in the simulation of a microwave heated fixed bed. The heat transfer equations, fluid dynamics and Maxwell Stefan equations for the electromagnetic field are solved simultaneously with the COMSOL Multiphysics software. **Figure 4d)** shows the simulation of the temperature evolution during microwave heating of a fixed zeolite bed together with the experimental results confirms a perfect temporal and spatial agreement.

In DIMAS a recent example of expertise and previous results and experience, applicants from the UPV, in collaboration with the CSIC, have patented a novel disruptive process of simultaneous activation of H_2O and green CO_2 by the application of microwaves that allows the production of green synthetic molecules with notable technological relevance. This development was published in the November 2020 issue of the prestigious journal Nature Energy[75] and has already been referenced in more than 100 scientific websites. The process uses electrical energy in the form of microwaves to carry out the non-thermal reduction of a catalytic solid metal oxide, releasing O_2, which is then reacted with water vapor and CO_2 to form syngas (a mixture of CO and H_2) that is used for the synthesis of sustainable organic molecules such as CH_4 or light olefins. Initial techno-economic studies of this process reveal that this technology can be competitive with other established technologies for converting electricity into green fuels.

4.6. Human resources, infrastructure and equipment available for the execution of the project

The **equipment at UNIZAR** includes own instruments available in NFP and GIFMA groups and also the access to the excellent facilities at the “Instituto de Nanociencia y Materiales de Aragón”, INMA and Advanced microscopy Laboratory (LMA) , a singular facility. All these instruments are located in the same building as the laboratories where the MW experimental set-up to develop this project.

Equipment in the laboratories of the NFP group:

Equipments for synthesis of materials and coating of surfaces

- Laboratory with basic equipment for synthesis of materials including stoves, autoclaves and ovens, calcination under different atmospheres and centrifuges.
- Equipments for coatings: Spray coating gun model SRI PRO LITE from DEVILVISS Dip coater KSV Nima

Characterization equipments

- Thermogravimetric analysis: Two equipments TA instruments TGAQ5000 equipped with autosampler and Mettler Toledo TGA/SDTA 851. Differential Scanning Calorimetry Mettler Toledo DSC 822e. The equipments will be used for thermal analysis of the prepared materials, i.e template removal and thermal stability.
- Dynamic light scattering DLS and zeta potential measurements (Brookhaven 90 Plus) will be used to evaluate particle size distribution and polydispersity of nanoparticles.
- Elemental analysis for the chemical composition of the solids prepared will be carried out by using Microwave Plasma-Atomic Emission Spectroscopy (MP-AES) Agilent 4100.
- The UV-VIS-NIR absorbance the LSPR of nanoparticles and the evolution of DMNP decomposition will be evaluated by using Agilent 8453 UV-Visible spectrophotometer and Varian UV-VIS 50 Probe.
- Nitrogen adsorption equipments ASAP 2020 and TRISTAR, (Micromeritics) for measurement of specific surface area, pore size, pore volume and pore size distribution of sorbents and catalysts.

Experimental set-up for MW heating

There are two existing set-up for MW heating: (i) 3-Stub Tuner monomode MW cavity able to operate at a maximum power of 150 W (manufacturer: Saica Iberica) in a TE310 resonant mode; (ii) a stainless-steel cylindrically-shaped resonator working in a TE111 resonant mode, where the maximum electric

[74] Julian, I.; Ramirez, H.; Hueso, J. L.; Mallada, R.; Santamaria, J. Non-Oxidative Methane Conversion in Microwave-Assisted Structured Reactors. Chem. Eng. J. 2018. https://doi.org/10.1016/j.cej.2018.08.150.

[75] J.M. Serra, J. F. Borrás-Morell, B. García-Baños, M. Balaguer, P. Plaza-González, J. Santos-Blasco, D. Catalán-Martínez, L. Navarrete and J. M. Catalá-Civera, "Hydrogen production via microwave-induced water splitting at low temperature". Nat Energy 5, 910–919 (2020). 10.1038/s41560-020-00720-6

field is found at the cavity center with a substantially constant field area (manufacturer: DIMAS group-Polytechnical University of Valencia). The setup also includes contactless thermometers: an infrared camera (Optris PI 1 M) and a pyrometer (Optris CTlaser LT). A gas chromatograph (GC) (Thermo Fisher Trace 1300 with auxiliary oven Trace 1310) in which 4 different columns and 3 detectors allow the identification and quantification of permanent gases (TCD), light hydrocarbons (FID1) and aromatics (FID2) in less than 10 min of analysis.

INMA and LMA facilities and equipment

-Transmission electron microscopy (TEM) 2 microscopes Tecnai (FEI) working at 200 and 300keV respectively. These microscopes allow performing selected area electron diffraction (SAED) and Electron energy loss spectroscopy (EELS). High-resolution TEM (HR-TEM) will be carried out with a FEI Titan 60-300, operating at 300 kV and equipped with an EDAX X-rays microanalysis.

-Scanning electron microscopy, SEM; two equipments FEI Inspect and environmental SEM for low conducting samples.

-Raman spectroscopy will be recorded with alpha300R WITEC Raman confocal microscope. The instrument is equipped with four different lasers, 488nm, 532nm, 633 and 785nm that could be selected as excitation source.

- Infrared spectroscopy FTIR. The equipment is a Vertex 70 Bruker (Bruker Optics Inc.) equipped with two detectors DTGS and MCD. This equipment will be important for characterization of surface functionalization of materials.

-X-Ray Diffraction, Empyrean from Panalytical, Sample compartments and accessories for the equipment include:spinner for reflection and transmission measurements with automatic charger.

-XPS X-Ray Photoelectron Spectroscopy, there are two equipments Ultra and AXIS SupraTM from Kratos, for surface analysis of the solids, chemical composition and oxidation state. Measurement range up to 2600eV.

Equipment in the laboratories of the GIFMA group:

- Coordinate Measuring Machine ZEISS PMC 876- CNC with a Vast continuous measurement probe Vast and the measuring software Calypso
- Coordinate Measuring Machine ZEISS PMC 850-CNC, with a continuous measurement probe and the measuring software METROLOG XG,
- High precision 3D printer OBJET EDEN 350V
- 3D Printer CubeX duo
- FDM 3D printer – Dynamical tools DT60
- 3D printer Sinterit Lisa Pro
- Focal variation microscope Alicona Infinite Focus SL, with the rotation unit IF-Real3DRotationUnit
- In addition, GIFMA research group has lead the scientific group of the application for the acquisition of an additive manufacturing machine for metallic components and a X-ray computer tomography (XCT) system for industrial applications (EQC2021-007097-P), which was granted in the call for scientific and technical equipment (EQC2021), for the SAI (research support service) of the University of Zaragoza. This equipment is due to be installed at the laboratory at the end of the year 2023.

DIMAS-UPV is well-equipped to push the boundaries of microwave research and technology transfer with one of the best-endowed microwave laboratories in Europe in this technology. The Microwave Laboratory is a state-of-the-art experimental facility that provides modern measurement capabilities in the field of high frequencies, microwave sensing, and microwave heating. It includes all the required equipment for the design of RF and microwave circuits, simulation, fabrication, measurement, and evaluation of prototypes or pilot plants. It accounts with more than 50 reactor systems designed and built in our state-of-the-art facilities: multi and single-mode reactors from few Watts up to 75KW in applications up to 1500ºC, with the possibility to perform simultaneous analysis of the materials under treatment through analysis techniques such as Raman, mass spectrometry or dielectric properties characterization. These reactors are among the most advanced in the world and are being extensively employed in contracts and collaborations. Some of the specific equipments together with a short description are listed below:

- Vector Network Analyzers (VNAs): High-quality microwave network analyzers for measurements of microwave parameters working up to 50GHz (Rohde&Schwarz models ZVA50, ZNB40, ZNB8-4 and ZVL 13)

- Microwave synthesizer Microwave equipment Anapico APSIN6010, from 9kHz to 6GHz, with 0.001Hz resolution and adjustable power (-120 ~ +17dBm).
- Mass spectrometer Equipment to identify chemical species (TecnoVac OmniStar), with qualitative and quantitative analysis (Detection range: 100% – 1 ppb)
- Raman Spectrometer Equipment with portable probe for in situ Raman spectroscopy ProRaman-L-532-B1S, TSI-Inc (Laser wavelength: 532 nm Power: up to 350mW).
- Laboratory fume cabinet Cabinet for gases control (Burdinola BST-1800), with 1800mm width and automatic flow control
- Temperature measurement instruments Temperature measurement instruments, including several pyrometers and thermal cameras, from -20°C up to 1500°C (Optris models PI400, PI1M, PI160 and FLIR ThermaCAM E45).
- High-power microwave generators Several 2.45GHz and 915MHz microwave high-power lab generators with adjustable power up to 6KW (2.45GHz) and 30KW (915MHz) (MKS models SM1280 & TM060, SM1250 & TM030, SM1250 & TM030, SAIREM model GLP180).
- Solid state microwave generators Several solid-state microwave generators with variable frequency signal and adjustable output power up to 600W (RF-Core models RCA2327H53B, RCAN2G4H56D1 and Sairem GLS600W); all at 915 MHz
- Measurement cells A large selection of measurement cells for broadband dielectric materials characterization (solids, liquids, powders, etc.).
- Additional electronic test instruments Spectrum analyzer Advantest R4131C, Digital oscilloscope GW instek GDS-840C, power meters (Anritsu ML2438A), etc.
- Precision Mechanical workshop Including CNC lathe (CMZ TB-46M) and milling machine (Lagun 4CNC)

5. EXPECTED IMPACT OF THE PROJECT

5.1. Expected impact in the generation of new scientific and technological knowledge

The results of the RASMIA project will represent a **significant advance in the increase in CO_2 capture rates accompanied by a reduction in the energy costs associated with capture**, which currently account for 70% of the cost of the amine-based adsorption process. Capture with solid adsorbents proposed in RASMIA requires less energy input for regeneration and they are also non-corrosive like amine-based liquid adsorbents.

The design of a structured S-G contactor with high packing density and low pressure drop is based on 3D printing technology. The monolith contains an adsorbent and/or catalytic material that can be heated with MWs for use in both a CO_2 capture system and a catalytic S-G reactor. **The development of structured adsorbents using customized and scalable additive manufacturing processes serves to create the basis and new knowledge in these sectors including catalysis with structured reactors**. Due to the low pressure drop with optimal gas-solid contact and packing, high flow rates can be processed resulting in **higher throughputs per unit volume, resulting in process intensification**.

Both the CO_2 desorption process and the methane reforming that is normally done with conventional heating (steam, combustion of fossil fuels), which uses inefficient heat transfer mechanisms based on conduction and convention, is improved with MWs heating. This is an **ultra-fast and volumetric heating that is more efficient,** where energy is transferred directly to the solid, **reducing energy consumption**. The **results of the project** for heating structured reactors with MWs **can be extended to other endothermic reaction systems requiring high reaction temperatures.**

The outcomes of RASMIA result in **improved CO_2 capture efficiency in combination with a novel regeneration technology, are easily extended to general gas cleaning using adsorption technology.** This research has a positive effect on the energy reduction for material heating in regeneration.

The project **will establish new scientific collaborations national and international with different sectors**. In general, the project will contribute knowledge in different research fields, solid materials for CO_2 capture, gas-solid contactors, and electrification of chemical processes. These aspects will be disseminated to the scientific community through publications in high impact index journals and the presentation of communications at international scientific conferences, as has been done on a regular basis by the project's research team.

5.2. Expected social and economic impact of the project

Social impact

At the recent United Nations Climate Change Conference 2021, COP-26 the European Commission supported the consensus reached by more than 190 countries to keep the Paris Agreement targets alive, aiming to limit global warming to 1.5°C. More specifically, the EU aims to reduce CO2 emissions from energy and energy-intensive industries by 80-95% by 2050. The microwave technology developed in RASMIA for CO_2 capture and use together with electrification of the chemical industry are recognized means to achieve this goal. CCUS is expected to play a key role in reducing emissions from the industrial sector, representing an emissions reduction contribution of 20-27% by 2060 (this corresponds to a cumulative value of 21-28 $GtCO_2$ or ~0.5$GtCO_2$/year).

Impact on job creation, competitiveness and employment

Pushing the boundaries of technology and innovation to meet the EU's 2050 climate and clean energy targets has already proven to be creating new industrial and job opportunities. 18 million net jobs can be created by 2030 worldwide by limiting global warming to 2°C by the end of the century[76]. A full implementation of the Paris Agreement could create 1.2 million additional jobs in the EU as a whole by 2030[77].

These job opportunities will be created along the value chain of the emerging low-carbon energy sector. According to the model, the deployment of CCS in Europe will create and secure an estimated total of 330,000 jobs in fuel supply, CCS equipment manufacturing, plant operation and operation of CO_2 storage facilities, while creating a completely new infrastructure for CO_2 transport and storage that can also be used by energy-intensive industries (e.g. steel, cement, refining, etc.)[78].

CCS clusters could also generate, both directly and indirectly, an economic impact totalling billions of euros from 2030 onwards, together with the potential to export this knowledge worldwide. Energy-intensive sectors (iron, steel, cement, refining, etc.) account for ~ 1.3 million jobs in the EU. More specifically, the steel industry directly employs more than 360000 people, which represents 1.25% of EU manufacturing employment and achieves an annual turnover of ~ ¤170000 million euros[79]. As this is a large sector, the deployment of the technology developed in RASMIA will create a significant number of jobs.

5.3. Technical dissemination strategy of the scientific results and internationalization plan

All partners will take an active role in discussing and deciding the content of the material to be published to ensure that no confidential information is disclosed which could jeopardize the industrial property or patentable techniques and technology resulting from the project. In all dissemination activities, it will always be considered that RASMIA technology can be applied across a wide range of manufacturing sectors. A communication strategy will be developed during the first months of the project that will detail the strategy of communication and dissemination, including the communication material, channels, target groups, timing, etc. **Table 4** shows the means already for dissemination. The strategy will be treated as a living document.

Table 4 *Means and dissemination strategy in RASMIA*

Means	Dissemination strategy
RASMIA project as part of the group's websites and the Institutions	Included on all communications to increase traffic. Highly ranked and linked from external sites. The most relevant results will be disseminated with the aid of the UNIZAR and UPV webpages, Twitter and relevant social media. yearly ITACA conference (http://www.itaca.upv.es/itaca-

[76] International Labour Organization ILO World Employment Social Outlook 2018. https://www.ilo.org/wcmsp5/groups/public/---dgreports/---dcomm/---publ/documents/publication/wcms_615594.pdf Accessed January 2023.

[77] European Commission. (2019) Sustainable growth for all: choices. for the future of Social Europe, Employment and Social Developments in Europe 2019. https://ec.europa.eu/social/main.jsp?catId=738&langId=en&pubId=8219 Accessed January 2023.

[78] https://zeroemissionsplatform.eu/wp-content/uploads/ZEPmarket-economics-report.pdf Accessed January 2023.

[79] https://zeroemissionsplatform.eu/wp-content/uploads/CCS-for-industry_-Lowest-cost-route-for-decarbonising-Europe.pdf

	wiict-2022/) called WIICT (Workshop on Innovation on Information and Communication Technologies)
Publications /Open access	Scientific publications in high-ranked peer-review journals from JCR including Chem. Eng. J., Appl.Catal.B, Angew.Chem, Chem. Eng. Proc. Proc. Intens. in addition to more specific journals in the field (IEEE Trans. on Microwave and Theory Techniques). We envision **at least 6 different publications in Open Access,** based on the publication agreement between the Universities and the main Publishers.
Press Releases	Creation of articles/ interviews for targeted media channels (radio, TV). Specific press releases prepared for project milestones and events.
Engaging Networks and Stakeholders	Communication of project activities and outcomes through networks – personal contacts, partner projects, international associations, consortia, European networks, Brokerage Days, Match-Making Info days, B2Match, etc (EUROPIC, ASPIRE, EFCE, OGMP)
Scientific events and conference participations	RASMIA public workshops and presentations at international conferences; establishing of sections oriented towards PI would increase the visibility and intensify the feedback from the community. Relevant international conferences will include the participation in the International Microwave Week, European Microwave Week, AMPERE Conference (International Conference on Microwave and High Frequency Heating), UCRA (Unconventional Catalysis and Reactors)

Regarding the Internationalization Plan, on medium to long-term substantial impacts to process engineering, unit process operations fields and catalysis can be expected. A corresponding transfer of the knowledge is one of the motivations for RASMIA. Another one is the close cooperation with the industry on advanced technology development. The main **technological innovations** are aimed at the application fields of apparatus design, process intensity and flexibility, tailored catalysts and process control. They integrate in an innovative way the advanced physical technologies and progress in the chemistry of catalysts to achieve: (i) improved control of the process ((ultra)short reaction time needed) direct heating via microwaves or radiofrequencies; (ii) New types of more robust catalysts less susceptible to surface deactivation and coke deposition suitable for the application in the novel process intensive applications under innovative operation regimes permitting a quick change of the product and reactions to changes in feed. Europe-wide dissemination and integration of the knowledge will be assured through the involvement of former partners in European and international scientific and technological organizations (**Table 5**).

Table 5. *Internationalization plan*

Organizations, Consortia, Groups, Stakeholders	**Potential Interests and partnerships**
The Oil & Gas Methane Partnership 2.0 (OGMP 2.0)	The Oil & Gas Methane Partnership 2.0 (https://www.ogmpartnership.com/ is a multi-stakeholder initiative launched by UNEP and the Climate and Clean Air Coalition.
A.SPIRE	A.SPIRE https://www.aspire2050.eu/aspire/the-association is the European Association which is committed to manage and implement the Processes4Planet co-programmed Partnership. It represents innovative process industries, 20% of the total European manufacturing sector in employment and turnover, and more than 180 industrial and research process stakeholders from more than 20 countries spread throughout Europe.

EUROPIC (the European Process Intensification Centre)	EUROPIC https://europic-centre.eu/ provides a channel for manufacturers, technology providers, equipment vendors and engineering contractors to share knowledge in order to realise the immense benefits process intensification brings.
Former collaborators	SAIREM's activity sector is Chemistry & Plasma industrial equipment; Johnson Matthey is a global supplier of catalysts, licensed technologies and other services to the petrochemical, syngas, oil refining and gas processing industries that participated in a former EU project (ADREM); DTI is a none-for-profit organization approved by the Ministry of Science providing industrial R&D support to around 10.000 customers a year. DTI develop new technologies ITACA has collaborations with University of Nottingham, whose Electromagnetic group, lead by Sam Kingman and Georgios A. Dimitrakis, develops microwaves devices for Industrial applications. And also "Püschner Microwave Power Systems" (https://www.pueschner.com/), a German company, that also constructs big microwave applciators for Chemistry, Pharma, Food, etc.
UNIZAR-UPV Internacionalization	UNIZAR and UPV participate national and international alliance of universities UNITA(https://univ-unita.eu/Sites/UNITA/en /) ENHANCE (https://enhanceuniversity.eu/) and Campus Rio Ebro to disseminate the most relevant results among close collaborating Universities at the National and European level.

5.4. Dissemination plan to target groups and society

The divulgation of the most relevant results to the public audience will be carried out through press releases, radio and TV interviews in public media. In addition, RASMIA will take advantage of other innovative strategies promoted by UNIZAR and UPV involving the Scientific Culture and Innovation Units from the network Fundación Española para la Ciencia y la Tecnología (FECYT) del Ministerio de Economía y Competitividad in coordination with the Image + Communication Gabinet from the Universities.

Table 6 *Dissemination activities*

Activity	**Description**
Dissemination Days	Early Stage Researchers and Experts in Scientific Divulgation share several days of conferences organized by the Universities
Women in Science	The International Day of the Women in Science provides visibility to the active role and experience of Women working in Scientific fields. Our team will participate in seminars in Secondary Schools providing a first-hand vision of the goals achieved in this project. Moreover, the visibility of Women in Engineering careers will be encouraged.
Dissemination in primary and secondary schools	Specific activities enrolled in collaboration programs with primary and secondary schools will be used to provide further insights about the topics and science covered in RASMIA.
Science Corner	A workshop is displayed once per month in a public food market in Zaragoza and different scientists take the opportunity to display a short survey of their research activity
European´s Night for Researchers	RASMIA will participate as part of UNIZAR and UPV to disseminate the most relevant results during the European´s Night for Researchers.
Monologues and PhD thesis threads	RISArchers https://ucc.unizar.es/actividad-divulgacion/risarchers merges fun + science with monologues performed by the researchers in multiple fórums (bars, workshops, conferences) and our aim is to prepare at least one or two monologues. PhD thesis threads will give the ESR the opportunity to briefly describe their PhD in short pitches.
TELECOCHIPS	TELECOCHIPS promotes the participatioin of High School students to rpomote the studies, and these event consist of 1.5 hours practical

	seminars involving research activities performed by lecturers at the School. It includes promotional material and short talks about the projects. We will organize a Microwave seminar to show the possibilities of the microwave engineering, including RASMIA results."

5.5. Transfer and valorization plan of the most remarkable results

The concept of solids regeneration with MWs is already described in the literature and published by researchers. Extending this concept of microwave regeneration in a structured bed with uniform heating and sufficient penetration of the MWs can generate a patent. The structuring of adsorbents with additive manufacturing technology has been studied in the last 5 years and is at an early stage. The concept of structured adsorbents specifically designed for adsorption of MWs has not been described and is therefore also eligible for protection. Once the preliminary results demonstrating the novel concepts of the project have been obtained, the appropriate strategy for the protection of the results will be studied with INMA and with OTRI offices at both institutions and experts in technology Identified stakeholders include former collaborators of both groups could be interested in the patent (SAIREM, Johnson Matthey and Püschner Microwave Power Systems).

5.6. Data Management Plan

The University of Zaragoza will make available to the PIs of the projects, through its management application, the generation of the Data Management Plan (DMP), following the instructions established by the Agencia Estatal de Investigación (AEI). Likewise, these data will be presented at the time they are required by the AEI and free repositories will be used (example Zenodo) for their deposit, dissemination and preservation.

All raw data and associated metadata will be curated in well-defined formats and stored for at least 10 years, ensuring the security of the experimental and simulation data. Access to these data will be initially restricted to the research team of this proposal. When there is no objection, data can also be made openly accessible earlier following a simple request. The results will be made available to the scientific public through peer-reviewed (open-access) publications, presentations at conferences, and storage in public databases.

5.7. Influence of the gender inclusion in the content of the proposal

This project is developed in the field of chemical engineering, specifically, it focuses on the development of CO_2 capture and utilization assisted by microwave heating. Due to the specific technological nature of the project, it has not been possible to include the gender dimension. However, it must be noted that in UNIZAR team, of the 5 researchers that participate in the project, 3 of them are women, including one of the main researchers of the project. This results in a high percentage of researchers who are women, considering the yet low representation of women in the technological sector.

6. BUDGET JUSTIFICATION

UNIZAR budget will be devoted to the following costs:

Personnel. A Postdoctoral Researcher will be hired for a period of 2 years to work mainly in the tasks devoted to the experimentation in the MW heating systems and in the operation of 3D printing. A person with a background in engineering and some experience dealing with experimental set up is required to perform the tasks described in WP4.

Travel expenses. Travel expenses to cover the attendance to international scientific meetings and/or workshops where the results will be divulgated. These costs will also include the coordination of joint meetings and experiments between the RASMIA members.

Equipment. During the lifetime of the project, it is necessary to maintain the existing equipment dedicated to the execution of the project, such as the GC and the MS mass spectrometer for online gas analysis. In addition, it also contributes to the maintenance of characterization equipment, used in part by this project, already existing in the research group. For the development of WP2 it is necessary to acquire a 3D printer + extruder, a viscometer for the ink formulation and a new laptop/computer to control the temperature loop.

Other goods and services (including consumables, small equipment). Estimated costs include the purchase of chemicals for synthesis, consumables for the upgrade of an experimental set-up for MWSA at UNIZAR and the construction of a cavity requiring brass and aluminium pieces together with

electronics, connectors and cables required for control loop, software licences, audit costs, registration fees for international meetings and joint experiments, glassware, monoliths, mortar, gas supply, GC columns, fittings, glassware, small equipment, stir plates, mass flow meters, IR optical thermometer or optical fibers for temperature measurement and control.

UPV budget includes costs in several items:
Personnel. We plan to contract one engineer for 2 years (2nd and 3rd) for performing measurements with the constructed applicators, and to collaborate in the required feedback to redesign cavities if needed.
Travel expenses. This project, and because of its nature, implies the participation in coordination meetings (between Zaragoza and Valencia) and also in conferences to publish the project results and also to facilitate the networking
Equipment: A *solid state source at 2.45GHz* is applied because the power control of this devices is much more accurate than the non-solid-state sources; and also, a simple *Climate Chamber* is required to perform measurements at controlled temperature to evaluate the effect of the temperature and humidity to the applicators and results. This will be specially relevant in the case of sorbent materials for MWSA, where properties may change dramatically with water at low temperatures. *Gas Crhomatograph,* micro-GC the DiMAS group has already a set-up for performing experiments with chemical reactions, include feed gas system that includes analysis with MS. In the case of DRM the product H2 can not be analyzed with MS and a GC is required.
Consumables: two kinds of consumables are considered: electronic, including the materials to manufacture the applicators, plus the chemical stuff to perform measurements, experiments, etc.
Other: in this field we consider several wastes like the licence software renewal, needed for the electromagnetic simulations, publishing costs, especially in open-access, maintenance of the workshop and the compulsory audit of the project. All these expenses must be covered by the project to success

7. TRAINING CAPACITY

7.1. Training Program in the context of the project

Individual research project and research skills for ESR1 at UNIZAR
The student will be incorporated in the doctoral program "Chemical Engineering and Environmental Technologies" that exists since 1987, now adapted to RD 99/2011. This program is one out of the 11 programs in the field of engineering and arquitecture offered by the Doctoral School of the University of Zaragoza.
The goal of his/her project is the development of a MWSA system for CO_2 capture with structured reactors prepared by additive manufacturing. In the methodology section, the WPs where the student will participate are indicated. Initially (M1 to M6) the student will acquire knowledge about the synthesis and characterization of materials. In M6 the student will start WP2 with simulation of fluid dynamics in COMSOL Multiphysics, and also deposition of materials in commercial monoliths. Once the simulations of at least one structure has been developed in M12 the student will start with the preparation of inks for printing and also in collaboration with GIFMA group she/he will learn about how to use 3D printer equipment. In M18 the student will start working on WP3 heating with microwaves the monoliths prepared in the MWSA set-up. At this point an iterative process will start based on the experimental results developing materials and monolithic structures for testing in MWSA. In M28 considering the best results obtained the student will perform the techno economic and CO_2 footprint of the process. After M30 the student will focus either on tests in the scale-up system in case the MWSA process has been selected for scaling or if not the student will look for technical aspects for improving the results on MWSA, other materials or structures.
The student will follow a safety course before entering the laboratory. The student will attend to specific courses on characterization and use of equipment regularly delivered in INMA and at the end of this training period the student will be able to manipulate in autonomous way: N2 adsorption, SEM/TEM, FTIR,TGA, UV-VIS, DLS, Raman.
*Training courses and soft skills***.** The student also can attend to different courses organized by the doctoral school and conferences in INMA, and other more specific about: "Intellectual property and know how", "Security and Classified Information" and "Ethics in research".

Individual research project for ESR2 at UPV

The OBJECTIVE of the Training Plan is to provide the pre-doctoral student with the necessary competencies for properly accomplishing his work functions, oriented towards scientific excellence.

The Phases of the proposed training plan are the following:

PHASE 1: Review of state of the art -> It takes place at the beginning of the project

PHASE 2: Performing simulations with ad-hoc and commercial simulators -> associated with task T2.2

PHASE 3: Design techniques to optimize applicators -> Associated with task T5.2

PHASE 4: Applicator construction and parallel tasks like control software, security systems, PID, ... -> associated with tasks T5.3 and T5.4

PHASE 5: Measurements, performing experiments, etc. -> Associated with tasks T1.3

PHASE 6: Publication of results -> Associated with the dissemination tasks

PHASE 7: Writing the Ph.D. thesis, review, and defense of the Ph.D. Thesis

In addition, the following actions are foreseen:

-Scientific writing workshop (CFP [Centro de Foramción de Postgrado] and ICE [Instituto de Ciencias de la Educación] at UPV offer seminars to the researchers to improve their results, and soft skills)

-The possibility of attending different courses organized by the doctoral school.

-Development in electromagnetic simulation skills.

Finally, the thesis will be directed by the PI of the project (Felipe L. Peñaranda Foix) and co-directed by José Manuel Catalá Civera.

Other common activities for ESR1 and ESR2 include:

Supervision and meetings. The student will meet on a regular basis, once a week, with the supervisors, IPs in this project for discussion and evaluation of the results and planning new experiments. On a monthly basis there will be an on-line meeting where the ESR1 and ESR2 will present their results to the whole consortium. Every six months in person meeting will be organized in the two institutions alternatively at UNIZAR and UPV, with the attendance of students.

In addition, the organization of NFP group includes a weekly group meeting (around 2 hours) where the Ph.D. students present their work in a structured way (formal PowerPoint presentations), and the results are discussed among all the group members. Our experience indicates that this type of meetings is highly beneficial for the students, improving their skills for scientific discussion, interpretation of results and planning of new experiments.

Communication skills. The group meetings will be a tool for communication skills but the student will also attend national/international conferences to present his own work. The student will learn on how to write a research article and should provide a scientific written report at the end of each year that will be evaluated by supervisors giving the corresponding feedback. This together with the preparation of manuscripts will be a training for the last period of the thesis devoted to writing PhD and defense.

Research secondments. The ESRs will perform secondments in international institutions, in particular:

ESR1 is expected to spend some time in the group of Dr. Marijke Jacobs in Flemish Institute for Technological Research, VITO in Belgium, they posses expertise in the field of 3D printing of porous materials and also either in the group of Prof. Andrzej Stankiewiz in Warsaw University of Technology (Poland) or Prof. Dionisos Vlachos in University of Delaware, both are experts in process intensification and also have experience in microwave heated chemical processes.

ESR 2 will stay in prestigious international centers which are regular collaborators of the DiMaS group. University of Nottignham (Faculty of Engineering -> Low Carbon Energy and Resources Technologies & George Green Institute for Electromagnetics Research), Universidad Politécnica de Cartagena (Grupo de Electromagnetismo y Materia), Università degli Studio di Modena e Reggio Emilia (Dipartimento di Ingegneria "Enzo Ferrari") and NIST (National Institute of Standards and Technology in Boulder-Colorado-USA)

7.2. PhD thesis defended and under supervision over the course of the past 10 years under the supervision of the research team

The thesis defended and under supervision, by UNIZAR and UPV together with the required information are presented in **Table 7**, where they are chronologically ordered, the thesis in bold are directly related to this project.

***Table 7**.PhD thesis defended and under supervision in the last 10 years by research team UNIZAR and UPV*

Inv	PhD candidate	Title	Starting date	PhD defense date	Publications
FPF & JMC	María Cabanes Sempere	Innovative production of nuclear fuel by microwave internal gelation.	13/10/10	19/07/13	Meas. Sci. Technol., 2013, 24, 095009; IEEE MTT-S International Microwave Symposium Digest (MTT), 2013, 1-4; IEEE/MTT-S International Microwave Symposium Digest, 2012, 1-3.
FPF	David Marqués Villarroya	Full wave circuit análisis of cylindrical elements for dieletrometry	25/07/14	14/09/18	IEEE Transactions on Microwave Theory and Techniques, 2018, 66(12); 48th European Microwave Conference, EuMC, 2018, ISBN: 978-2-87487-051-4; IEEE Transactions on Microwave Theory and Techniques, 2017, 65(4); Progress In Electromagnetics Research M, 2017, 58, 1-10; 47th European Microwave Conference, EuMC, 2017, ISBN: 978-2-87487-047-7.
JMC	Pedro. J. Plaza González	**Control de la temperatura en sistemas de calentamiento por microondas**	21/12/99	14/01/16	Nat. Energy, 2020, 5, 910-919; IEEE Transactions on Instrumentation and Measurement, 2020, 69(6), 3595-3605.
RM	Hakan Nigar	**Microwave Heating in Adsorption Based Pollutant Control Systems**	1/05/13	14/07/17	Sep. Purif. Technol. 151, 193–200 (2015); AIChE J. 62, 547–555 (2016); Environ. Sci. Technol. 52, 5892–5901 (2018); Chem. Rec. 19, 40–50 (2019); Appl. Therm. Eng. 155, 226–238 (2019).
RM	Adrian Ramirez	**Reactores catalíticos gas-sólido con calentamiento directo del sólido en campo microondas.**	10/01/14	20/10/17	Angew. Chemie - Int. Ed. 55, 11158–11161 (2016); Mater. Res. Express 1, (2015); Appl. Catal. A Gen. 505, 193–199 (2015); Catal. Today 210, 89–97 (2013); Catal. Today 273, 99–105 (2016); ChemistrySelect 2, 8509–8516 (2017); Chem. Eng. J. 316, 50–60 (2017); Sci. Adv. 5, (2019); Chem. Eng. J. 393, 124746 (2020);
RM	Angela Lopez	Estudio y optimización de catalizadores bimetálicos. Proceso de Oxidación Preferencial de CO	15/01/13	03/11/17	Applied Catalysis A: General, 2016, vol 528, pp. 86-92 (2016); J. of Membrane Science, vol. 523, p. 24-35 (2017),. Frontiers in Chemistry, 2018, 6,487 (2018)
JLH	M.Carmen Ortega-Liebana	Light-emitting carbon nanodots. Synthesis and development of novel applications	10/01/14	05/09/18	Catalysis Communications 2014, 56, 115-118; JPPC 2015, 119 (4), 2158-2165; Chemical Communications 2015, 51 (93), 16625-16628; Diamond and Related Materials 2016, 65, 176-182; Applied Catalysis B-Environmental 2017, 218, 68-79; ACS Applied Materials & Interfaces 2017, 9 (47), 41529-41536; Carbon

					2017, 117, 437-446; Chemistry-A 8European Journal 2017, 23 (13), 3067-3073; Nanoscale 2017, 9 (5), 1787-1792; Chemical Science 2018, 9 (40), 7766-7778; Chemistry-A European Journal 2019, 25 (21), 5539-5546; Carbon 2020, 156, 453-462; Catalysts 2020, 10 (3)
RM	Mariella Polino	Ion-exchange membranes for protein crystallization and protein crystals	1/09/14	30/04/19	Cryst. Growth Des. 17, 4563–4572 (2017); Cryst. Growth Des. 19, 4871–4883 (2019); Cryst. Growth Des. 20, 2174-2186 (2020); Membranes 11, 549, (2021).
FPF & JMC	Younes Ahajjam	Design and Implementation of a UWB Radar Sensor for Non-Destructive Application	27/09/13	11/06/19	Advanced Electromagnetics, 2019, 8(3), 76–82; 5th International Conference on Multimedia Computing and Systems (ICMCS), 2016, pp. 497-501, ISBN: 978-1-5090-5146-5; MATEC Web Conf., 2018, 191, 00002; Procedia Manufacturing, 2019, 32, 710-716.
RM	Diego Sanz	Desarrollo de fotocatalizadores activos bajo luz visible para la degradación de COVs	1/10/14	18/11/19	J. Hazard. Mater. 384, art. no. 121279 (2020).
RM	Marta Lafuente	Estrategias para la detección de compuestos organofosforados en fase gas y líquida mediante espectroscopía Raman-SERS	1/02/16	11/11/20	Micromachines 9(2), 60-65, (2018); Sensors Actuators, B Chem. 267, 457–466 (2018); Nanoscale 11, 12152–12160 (2019), J. Hazard. Mater. 384, art. no. 121279 (2020); ACS Applied Materials and Interfaces, 12 (32), pp. 36458-36467, (2020); Applied Surface Science, 506, art. no. 144663, (2020). ACS Sensors, 6 (6), pp. 2241-2251(2021)
RM	Roberta Manno	Microwave-Assisted Synthesis of Supported Nanocatalysts: A route from nanoparticles to nanoclusters, from batch to continuous.	1/03/17	12/02/21	Ind. Eng. Chem. Res., 58 (28), pp. 12711-12711, (2019); Chemistry of Materials, 32 (7), pp. 2874-2883, (2020); Catal. Today, 362, pp. 81-89, (2021); Nanoscale Advances, 3 (4), pp. 1087-1095 (2021) Green Chemistry, 22(4), pp. 1414–1423, (2020);
PLG	Alberto Clemente Cornago	Nuevas estrategias para la monitorización de nanomateriales en entornos de interés ocupacional y medioambiental.	01/01/13	16/06/17	Aerosol Sci. and Tech., 2013, 47 (8) 867-874; J. Haz. Mater. 2014, 280, 226-234; Env. Sci. Nano, 2016, 3(3) 496-503; Env. Sci. Nano, 2017, 4, 631-637; Journal of Colloid and Interface Science, 513 (2018) 497-504; Science of the Total Environment. Volume 625, (2018), Pages 978-986; Journal of Hazardous Materials, 363 (2019) 358-365.

JMC	José D. Gutiérrez Cano	Microwave Dielectrometry Adapted to Environments	19/09/16	19/07/22	Sensors, 2021, 21(16), 5463; IEEE Microwave and Wireless Components Letters, 2021, 31(6), 624-627; Journal of Food Engineering, 2022, 316, 110823; IEEE Transactions on Geoscience and Remote Sensing, 2020, 58 (5), 3097-3112; IEEE Transactions on Instrumentation and Measurement, 2020, 69(6), 3595-3605; IEEE Transactions on Microwave Theory and Techniques, 2017, 65(4), 1191-1202; Journal of Food Engineering, 2017, 202, 1-8; Progress In Electromagnetics Research M, 2017, 58, 1-10; 47th European Microwave Conference (EuMC), 2017, 440-443.
PLG	Beatriz Arauzo Gimeno	Aerosoles biocompatibles a partir de polvo seco: Nuevo enfoque en el suministro de fármacos por vía inhalatoria.	01/06/18	27/09/22	Pharmaceutics 2022, 14(12), 2763; Carbohydrate Polymers, 2021, 273, 118543; Pharmaceutics 2021, 13, no.12: 1988; Materials 2021, 14(24), 7601
RM	Cristobal Carrillo	Adsorción y catálisis para la descontaminación de agentes de guerra química	16/10/18	Expected May 2023	
JLH	Javier Bonet Aleta	Síntesis, desarrollo y transporte selectivo de catalizadores nanoestructurados para la alteración metabólica y destrucción de células cancerígenas	20/09/19	Expected Sept 2023	Catalysts 2020, 10 (3); Catalysts 2020, 10 (12); Catalysts 2022, 12 (2); Nanomaterials 2022, 12 (5); Journal of Colloid and Interface Science 2022, 617, 704-717; Chemical Science 2022, 13 (28), 8307-8320; Applied Materials Today 2022, 29; Chemical Society Reviews 2022, 51 (17), 7662-7681; Materials Today Chemistry 2023 (accepted)
JLH	Jose Ignacio Garcia Peiro	Síntesis, desarrollo y transporte selectivo de catalizadores nanoestructurados para la alteración metabólica y destrucción de células tumorales	01/12/19	Expected Dec 2023	Catalysts 2020, 10 (12); Catalysts 2022, 12 (2); Nanomaterials 2022, 12 (5); Chemical Society Reviews 2022, 51 (17), 7662-7681
PLG	Sergio Aina	Desarrollo de nanomateriales para aplicaciones en energía	01/08/20	Expected Dec 2023	Electrochimica Acta. Volume 424, 20, 140649; Sustainable Material Solutions for Solar Energy Technologies, 2021. Elsevier, ISBN: 9780128215920.

		(almacenamiento y generación)			
PLG	Raul Boix	Desarrollo de nuevos catalizadores heterogéneos fotoactivos	31/10/20	Expected May 2024	
FHA	Ángeles Valls Chiva	Síntesis, caracterización y vehiculización de nanomateriales catalíticos destinados a alterar el microambiente tumoral.	27/10/20	Expected Oct 2024	
RM	Kissia Batista	Advanced Surface Enhanced Raman Spectroscopy (SERS) for gas and liquids sensING in the area of chemical protection	4/03/21		
RM	Mª Isabel Sanchez	Respuesta efectiva ante amenazas de riesgo química combinando tecnologías basadas en SERS y catálisis	15/10/21		
RM	Javier Midón	Desarrollo de fibras, polímeros y cerámicas con propiedades foto-catalíticas intrínsecas	15/10/21		

7.3. Scientific and professional trajectory by the PhD

The actual positions of the 10 PhDs awarded at UNIZAR include: 4 postdoctoral research positions in international well recognized Universities: Delft, ETH-Zurich, Twente, Edinburg; 4 postdoctoral positions in Spanish National Research Centers and Universities: U. Publica Navarra, National Renewable Energy Centre (CENER), GENYO-U. Granada; 2 R+D contracts as analysts/consultants, consultants; 1 position in Secondary H.S. Teaching and the latest one searching for offers in the Pharmacy Industry. It is also worth mentioning an associated editorial position in Elsevier and the awarding of several postdoc positions in competitive calls: M. Zambrano, M. Salas, M. Curie.
The actual positions of the 5 PhDs awarded at UPV include: 2 postdoctoral positions at U.P. Valencia, 2 positions at technological companies Product Manage and Information Architect: Hiberius, Duagon and 1 position as Secondary School lecturer.